EndPCA: Ensemble Defense With Provably Convergent Aggregation Against Poisoning Attacks in Federated Learning

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***Abstract*—Despite its success in many applications, federated learning is increasingly vulnerable to sophisticated poisoning attacks. Existing defenses, particularly Byzantine Robust Aggregation Rules (BRARs), offer some protection but rely on strong assumptions or challenging technical prerequisites. To address these shortcomings, we propose an *ensemble defense with provably convergent aggregation* (EndPCA). By using the entropy weight method to consolidate scores from multiple BRARs into an ensemble trust score, it effectively integrates heterogeneous weak BRARs to resist a wide range of poisoning attacks under practical assumptions. We formally prove that EndPCA can provide theoretical guarantees of convergence with bounded error. Our empirical evaluations show that EndPCA consistently outperforms existing BRARs, demonstrating its effectiveness across various scenarios.**

***Index Terms*—Federated learning, poisoning attacks, byzantine robust aggregation rule.**

I. Introduction

FEDERATED learning (FL) enables decentralized client devices to train machine learning models while adhering to data privacy constraints. In FL, the server distributes a global model to clients, which train it locally on private data and return updates. The server then aggregates these updates to refine the shared model iteratively until satisfactory accuracy is achieved. FL offers secure on-device learning and has been deployed across healthcare [1], finance [2], autonomous vehicles [3], and Industry 4.0 [4], [5]. It is already used by companies like Google and Apple in services like Gboard's next-word prediction [6] and Siri [7].

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Although FL is valuable, many of its application scenarios, exemplified by privacy-sensitive sensing, adversarial manipulation and blockchain environments with decentralized trust and immutable but contamination-prone data [8], [9], [10], inherently expose models to inference and perturbation risks. These characteristics suggest that FL systems are increasingly vulnerable to sophisticated and evolving attacks. This work focuses on defending against *poisoning attacks*, which are considered among the most harmful threats to FL [11], [12], [13]. Poisoning attacks come in two main types: *untargeted attacks*, aiming to reduce the accuracy of the global model [14], [15], and *targeted attacks*, aiming to change the model's predictions for specific inputs while leaving others unchanged [16], [17]. For instance, targeted attacks might involve inserting *backdoors* [16], [18] into the global model, allowing adversaries to control predictions for predefined inputs without triggering suspicion during normal operations.

Byzantine Robust Aggregation Rules (BRARs) are effective against poisoning attacks [19], [20], [21]. BRARs apply various techniques to analyze the model updates uploaded by clients, mitigating the negative impact of the poisoned updates on the global model. We categorize BRARs into three types: robust statistical-based methods [19], [21], [22], methods using an auxiliary trusted dataset for evaluation [23], [24], [25], and hybrid approaches combining both [15], [26], [27]. They rely on strong (and often impractical) assumptions or technical prerequisites that are challenging to implement in practice. Examples include requiring a certain proportion of honest clients [21], specific properties of honest updates [28], or a clean and unbiased dataset [26]. Deviations from these assumptions or technical prerequisites can significantly undermine their effectiveness. Therefore, although they may offer robust defenses in specific attack scenarios, they fall short in defending against a wide range of poisoning attacks, particularly unknown ones.

We observe that existing BRARs exhibit complementary strengths: different BRARs excel in specific defensive scenarios, while their weaknesses are often covered by the strengths of other BRARs. FLTrust [26] performs well in independent and identically distributed (IID) but struggles in non-IID settings. FLAME [29] is effective in both, but weakens when malicious clients exceed half. The key question we address is whether a unified framework can integrate complementary BRARs while ensuring provably convergent robust aggregation under diverse and dynamic adversarial conditions.

Our Works. To address the above question, we develop *Ensemble defense with Provably Convergent Aggregation*

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(EndPCA), a unified theoretical framework that adaptively integrates multiple base BRARs, each tailored to specific attack patterns and FL conditions, into a robust and convergent aggregation scheme. Integrating these base BRARs is a highly technical challenge. Essentially, it involves combining multiple partial and potentially conflicting assessments of the same entity into a comprehensive assessment. In particular, a naive combination (e.g., averaging) of base BRARs often weakens the overall defense capability. We tackle this challenge by leveraging a core idea of ensemble learning [30], [31], i.e., multiple weak learners can be combined into a stronger learner. The design and implementation of EndPCA incorporate the following key techniques. (1) We develop an *entropy-weight-method-based integration*, which calculates the entropy of the trust scores assigned by BRARs to model updates and dynamically adjusts an importance score for each base BRAR. (2) We select a compact yet diverse set of base BRARs to cover a wide range of poisoning attacks and FL scenarios, while keeping the system complexity low. To enable seamless integration, we adapt existing BRARs with key modifications, such as constructing trusted datasets from client-side data and using clustering methods to select robust reference points.

EndPCA not only strengthens protection against poisoning attacks that the original base BRARs could counter but also extends defense to attacks they couldn't. Importantly, one of our key design choices is to construct trusted datasets from client-side data instead of relying on a server-side trusted dataset. We extend the convergence proof of FLtrust [26] to this setting, showing that EndPCA achieves convergence without requiring an auxiliary trusted dataset, which represents a key innovation not supported by FLtrust. Finally, extensive experiments demonstrate that EndPCA significantly outperforms strong baselines, including state-of-the-art (SOTA) BRARs, achieving 0.02% to 7.22% higher accuracy and 0.04% to 74.02% lower attack success rates.

Contributions. We make the following contributions:

- We propose a unified defense framework, capable of robustly integrating heterogeneous weak BRARs, via a tailored ensemble scoring mechanism based on entropy.
- We adapt complementary BRARs to instantiate EndPCA, not by simply combining them, but by bridging their differences through targeted modifications. These include constructing client-side trusted datasets and using clustering to select robust references, enabling an efficient and unified defense across diverse poisoning scenarios.
- We formally prove that EndPCA can provide theoretical guarantees of convergence without an auxiliary trusted dataset, making it more broadly applicable to practical FL systems than existing BRARs.
- We empirically evaluate the effectiveness of EndPCA. The results show that it outperforms existing BRARs in both metrics related to robustness and backdoor resilience.

II. Background and Related Work

A. Background

Following the previous works [19], [26], [32], we consider a typical FL system consisting of a centralized server and n clients labeled by $1, \ldots, n$. The set of all client numbers is denoted as $\mathbb{N} = \{1, \ldots, n\}$. The server manages a global model, with parameters denoted as $\boldsymbol{w}_G$. Each client holds a local private training dataset D_i, where $i \in \mathbb{N}$, exclusively accessible to client i; other clients and the server cannot access to D_i. The joint training dataset of all clients is denoted as $D = \bigcup_{i=1}^{n} D_i$. The training example in D follow an unknown distribution $\mathcal{X}$. Each training example in D_i follows a unique distribution (i.e., non-IID). Each client manages a local model, with parameters denoted as $\boldsymbol{w}_i$, where $i \in \mathbb{N}$. Formally, the goal of FL is to solve the optimization problem, $\boldsymbol{w}^* = \arg\min_{\boldsymbol{w} \in \mathbb{R}^d} \mathcal{L}(\boldsymbol{w})$, where $\mathcal{L}(\boldsymbol{w}) = \mathbb{E}_{X \sim \mathcal{X}}[L(X, \boldsymbol{w})]$ is the expectation of the empirical loss $L(X, \boldsymbol{w})$ on distribution $\mathcal{X}$, $\mathbb{R}^d$ is the parameter space, and d is the parameter dimensionality. As illustrated in Fig. 1, FL system trains the global model through multiple training iterations, and performs the following steps at each iteration. (1) At t^{th} iteration, the server sends the current global model parameter $\boldsymbol{w}^t{}_G$ to the clients, denoted as $\mathbb{S}^t \subset \mathbb{N}$ (①). (2) Each client learns a local model parameter $\boldsymbol{w}^t{}_i$ by solving the optimization problem $\boldsymbol{w}^t{}_i = \arg\min_{\boldsymbol{w}} \sum_{X \in D_i} L(X, \boldsymbol{w})$. Client i initializes its local model with $\boldsymbol{w}^t{}_G$, and performs several iterations via stochastic gradient descent based on D_i (②). Each client sends its local model update $\boldsymbol{g}_i^t = \boldsymbol{w}^t{}_i - \boldsymbol{w}^t{}_G$ to the server (③ and ④). (3) The server aggregates these local model updates $\{\boldsymbol{g}_i^t : i \in \mathbb{S}^t\}$ into a global model update $\boldsymbol{g}_G^t$. The aggregation function is $\mathbb{G} : \mathbb{R}^{|\mathbb{S}^t| \times d} \to \mathbb{R}^d$. The server then updates the global model parameters with $\boldsymbol{w}^t{}_G = \boldsymbol{w}^{t-1}{}_G - \alpha \boldsymbol{g}_G^t$, where $\alpha \in [0, 1]$ is the global learning rate. The superscript t in symbols like $\boldsymbol{g}_G^t$ and $\boldsymbol{w}^t{}_G$ represents the global training iteration. Unless involving training iterations, the superscript is omitted in the following sections.

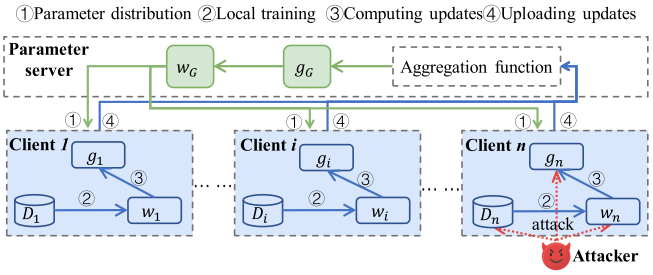

Fig. 1. The training process and the poisoning attack in FL.

B. Poisoning Attacks

As illustrated in Fig. 1, attackers control malicious clients to send harmful local model updates to the server with the intent of corrupting the global model. These malicious updates may be generated from poisoned local training data [33], [34], [35] or entirely fabricated [14], [36], [37]. Poisoning attacks can be classified into *untargeted attacks* and *targeted attacks*. In untargeted attacks, the attacker aims to broadly degrade the global model's accuracy. Examples include the Labelflip [38], Trim [14], and adaptive attack [14]. Targeted attacks seek to manipulate the model's prediction on specific inputs while preserving normal behavior elsewhere. A prominent example of a targeted poisoning attack is the backdoor attack [16], [18], [36]. Here, the attacker embeds a trigger, such as a unique pattern or signal, into target test inputs to compromise the model, forcing it to misclassify or favor specific outcomes. The typical backdoor attacks are 3DFed [17] and scaling attack [16].

C. Byzantine Robust Aggregation Rule

BRARs defend against poisoning attacks, enabling servers to learn accurate global models despite Byzantine clients. They

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fall into three categories. The first one uses robust statistical methods, treating model updates or gradients as vectors and mitigating the impact of statistical outliers. The second one uses a trusted dataset to evaluate model performance, assess client trust, and assign aggregation weights. The third one combines statistical methods with the trusted dataset. In the first methods, Foolsgold [28] computes the similarity between historical as an indicator of potential attacks. Deepsight [19] proposes division differences and normalized energy updates to differentiate between malicious clients and benign ones. RFLBAT [39] employs K-means for aggregation. FLAME [29] uses clipping, clustering and noise to mitigate malicious updates. In the second methods, out-of-distribution (OOD) data is used for detecting backdoors [40]. FAIR [41] estimates learning quality using historical records. Zeno [25] collects a validation dataset and calculates validation gradients. FLDetector [24] excludes outliers through comparing historical updates with the local updates from the current epoch. In the third methods, FLTrust [26] assigns each client a score based on the similarity between the model updates trained on the root dataset and the local model updates. FLest [42] enhances FLTrust by clustering the clients' updates. FLCert [43] integrates multiple BRARs based on the principle of majority consensus.

However, existing BRARs still face notable limitations. Firstly, many rely on overly strong assumptions about attackers and training data. For example, FoolsGold [28] holds diversity assumption for the honest updates. Krum [22], Bulyan [21], and Trimmed mean [44] require the proportion of malicious clients to stay below strict thresholds. Secondly, some BRARs rely on technical prerequisites that are challenging to implement. For example, FLTrust [26] assumes access to a clean and unbiased trusted dataset. From the perspective of provable security, Bulyan [21] and FLCert [43] can guarantee convergence only under strict limits on the number of malicious clients. Currently, no BRAR can address **both targeted and untargeted attacks under conditions where the trusted dataset may be biased, or malicious clients may exceed half, while also providing theoretical guarantees**.

III. Threat Model and Requirements

A. Threat Model

We adopt the threat model used in previous works [14], [26], [45] on poisoning attacks. The attacker's goal is to execute either untargeted or targeted poisoning attacks to undermine the FL training process. We assume that an attacker can control at most m malicious clients, where $m < n - 1$, implying that there is at least one honest client. The attacker can instruct these malicious clients to execute designated attacking tasks, including submitting arbitrary model updates to the server. The server is semi-honest and trusted in computation, while clients may be malicious but cannot modify other clients' data or the server's logic. The attacker selects malicious clients before training but cannot manipulate originally honest clients during the training process, with client identities (malicious or honest) remaining unchanged. Meanwhile, the attacker may exploit its background knowledge of the target FL system to launch attacks. We focus on the strong poisoning attacks in the full-knowledge setting [26], [46], which assumes the attacker knows the global model, the global model updates, the global learning rate, the server's aggregation rule, the loss function, the number of all clients, the local training data of malicious clients, and the local model updates on all clients.

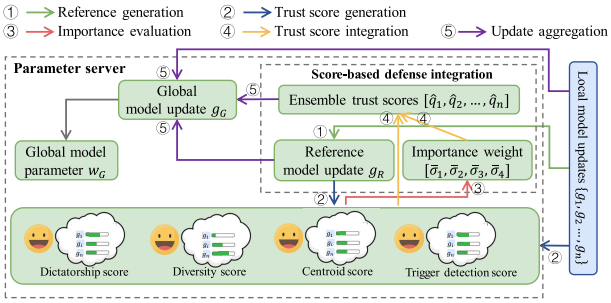

Fig. 2. Overview of EndPCA.

B. Design Goals and Requirements

We aim to design a broad-spectrum FL defense that meets key security requirements: (R1) **Defense-integration robustness**: The defense should resist a broad-spectrum of targeted and untargeted poisoning attacks, preserving the global model's accuracy and backdoor resilience as if no attacks occurred. (R2) **Fidelity**: The method should preserve the global model's classification accuracy under no attacks, matching the performance of FedAvg [32]. (R3) **Data distribution independence**: The defense should work effectively for both i.i.d. and non-i.i.d. client data distributions, without relying on specific distribution characteristics. (R4) **Large-scale malicious clients resilience**: The method must remain robust against a wide range of malicious client proportions, performing reliably even when the proportion of honest clients falls below 10%. (R5) **No-Trusted server-side datasets**: The method avoids requiring a trusted dataset on server, relying solely on client-uploaded information to establish trust and guide model aggregation.

IV. The Proposed EndPCA

A. Overview of EndPCA

We propose a defense integration method that unifies multiple heterogeneous strategies against diverse poisoning attacks, while remaining easily extensible. Since naive combinations (e.g., averaging) often weaken individual defenses, we design an *entropy-weight-method-based integration* that merges their trust scores into a single *ensemble trust score* to preserve and enhance their effectiveness. The ensemble score, weighted by the entropy of defense trust scores (lower entropy contributes more), serves as the aggregation weight in $\mathbb{G}$. The method also computes a reference model update as the trust root. We select suitable base BRARs based on the following principles. (1) Dictatorship, i.e., computing one of the most "promising" model updates to judge other updates. (2) Diversity, i.e., distinguishing the honest clients and the malicious ones based on the diversity of their updates. (3) Centroid, i.e., clustering the model updates to aggregate similar ones into a "high-promising" updates. (4) Trigger detection, i.e., identifying changes in specific neurons to determine if a backdoor is implanted. While no single principle meets all five requirements.

Fig. 2 provides an overview of EndPCA, where the server employs four scoring defense strategies derived from SOTA defense or attack techniques and tailored for integration based on the four defense principles. At each global training iteration, a reference model update is computed based on the local model updates from $\mathbb{S}^t$ (①). Then, the four defense

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TABLE I
CAPABILITY COMPARISON OF DEFENSE METHODS

Method	R1	R2	R3	R4	R5
FLTrust [26]	×	✓	×	✓	✓
FLest [42]	×	✓	×	×	✓
Foolsgold [28]	×	✓	✓	×	✓
FLDetector [24]	×	✓	✓	×	×
FLAME [29]	×	✓	✓	×	×
RFLBAT [39]	×	✓	✓	✓	×
Deepsight [19]	×	✓	✓	✓	×
FLCert [43]	✓	✓	×	×	×
EndPCA (ours)	✓	✓	✓	✓	✓

strategies assess local model updates, assigning trust scores (Dictatorship, Diversity, Centroid, and Trigger detection) to clients (②). Next, the importance weight for each defense strategy is calculated based on the entropy of its generated trust scores (③). Ensemble trust scores are computed as the weighted sum of trust scores under the importance weights (④). Finally, the global model update is obtained through weighted averaging of local updates, adjusted by ensemble trust scores and normalized with reference model updates (⑤).

We compare the capabilities of various BRARs in Table I. EndPCA satisfies all the security requirements, while no other algorithm meets them all. It defends against the most poisoning attacks among all methods, performing well across five key defense capabilities. We emphasize that EndPCA does not rely on the idea that incorporating more defenses necessarily leads to better performance. The framework adopts four components because they correspond to the four principles that jointly satisfy all required security requirements (R1-R5). Introducing additional BRARs yields limited marginal benefit unless new security objectives need to be addressed.

B. Entropy-Weight-Method-Based Integration

Prior work [15], [26], [28] suggests that effective scoring-based defenses yield polarized trust scores. This aligns with the principle of the entropy weight method (EWM) [47], [48], which assigns higher weights to low-entropy (i.e., more informative) metrics. Empirically, we observe that more robust methods consistently exhibit lower entropy in trust scores (refer to Section V-E), supporting the use of EWM in computing ensemble trust. At t^{th} global training iteration, i^{th} defense strategy computes a set of trust scores for the selected clients, i.e., $\{\boldsymbol{c}_{i,j}{}^t\}_{j\in\mathbb{S}^t}$, where $\boldsymbol{c}_{i,j}{}^t$ is the trust score given by i^{th} defense strategy to client j at t^{th} iteration, and the method for computing $\boldsymbol{c}_{i,j}{}^t$ will be detailed in Section IV-C. To ensure fair weight calculation across defense strategies, we normalize $\boldsymbol{c}_{i,j}{}^t$: $\boldsymbol{c}_{i,j}{}^{t,\text{norm}} = \boldsymbol{c}_{i,j}{}^t / \sum_{j\in\mathbb{S}^t} \boldsymbol{c}_{i,j}{}^t$. Entropy reflects the dispersion of each defense's scores:

$$e_i = -\frac{1}{\ln(|\mathbb{S}^t|)} \sum_{j\in\mathbb{S}^t} \boldsymbol{c}_{i,j}{}^{t,\text{norm}} \ln(\boldsymbol{c}_{i,j}{}^{t,\text{norm}} + \epsilon_{ent}) \tag{1}$$

where ϵ_{ent} is a small value to prevent ln(0). The importance weight $\bar{\sigma}_i$ of defense i is derived from entropy value. Defenses with lower entropy are assigned higher weights:

$$\bar{\sigma}_i = \frac{1 - e_i}{\sum_{i=1}^{K}(1 - e_i)} \tag{2}$$

where K is the number of all defense strategies.

To capture the relative position of trust scores on a global scale, we calculate the fuzzy membership degree for each score: $\boldsymbol{c}_{i,j}{}^{t,\text{fuzzy}} = (\boldsymbol{c}_{i,j}{}^t - \min_{j\in\mathbb{S}^t} \boldsymbol{c}_{i,j}{}^t)/(\max_{j\in\mathbb{S}^t} \boldsymbol{c}_{i,j}{}^t - \boldsymbol{c}_{i,j}{}^t)$. The aggregated score for client i is a weighted sum of fuzzy membership values: $q_j^t = \sum_{i=1}^{K} \bar{\sigma}_i \boldsymbol{c}_{i,j}{}^{t,\text{fuzzy}}$. To address the potential randomness in each iteration, we design a historical weighted average formula to integrate current and historical scores into an ensemble trust score $\hat{q}_i^t$:

$$\hat{q}_i^t = \hat{q}_i^{t-1} + \alpha_q(|q_i^t - \hat{q}_i^{t-1}|)(q_i^t - \hat{q}_i^{t-1}) \tag{3}$$

where $\alpha_q(x) = \max\{1, 2x - x^2\}$ is used to balance the historical scores and the current scores. The reference model update for the $(t+1)^{\text{th}}$ iteration is computed as:

$$\boldsymbol{g}_R^{t+1} = \frac{\boldsymbol{g}_k^{t+1}}{t+1} + \frac{t\boldsymbol{g}_R^t}{t+1}, \quad \text{s.t. } k = \arg\max_{i\in\mathbb{S}^t} \hat{q}_i^t. \tag{4}$$

It iteratively computes the average model update with the highest $\hat{q}_i^t$ in history. Specifically, $\boldsymbol{g}_R^{t+1} \approx \frac{1}{|\mathbb{S}_R^t|} \sum_{i\in\mathbb{S}_R^t} \boldsymbol{g}_i^t$, where $\mathbb{S}_R^t$ includes clients with the highest $\hat{q}_i^t$ from the initial to t^{th} iteration. A larger $|\mathbb{S}_R^t|$, excluding malicious updates, reduces $\|\boldsymbol{w}^* - \boldsymbol{w}^t{}_G\|$, thereby improving global model quality. See Section IV-E for theoretical analysis. Even if early camouflaged attackers briefly enter the reference set, their contribution is suppressed by the 1/t-decaying historical averaging in Eq. (4), making such contamination quickly negligible in later rounds.

The aggregation function $\mathbb{G}$ is defined as follows:

$$\boldsymbol{g}_G^t = \frac{1}{\sum_{j\in\mathbb{S}^t} \hat{q}_j^t} \sum_{i\in\mathbb{S}^t} \hat{q}_i^t \cdot \frac{\|\boldsymbol{g}_R^t\|}{\|\boldsymbol{g}_i^t\|} \cdot \boldsymbol{g}_i^t \tag{5}$$

where term $\frac{\|\boldsymbol{g}_R^t\|}{\|\boldsymbol{g}_i^t\|}$ normalizes $\boldsymbol{g}_i^t$, constraining its magnitude to $\|\boldsymbol{g}_R^t\|$ while preserving its direction. This prevents anomalous updates from affecting the aggregation. Theoretical properties of this normalization are key to the proof of Lemma 1, which is a crucial step in proving the convergence. The EWM-based integration also behaves stably under two boundary conditions. When one sub-defense produces highly polarized scores, its entropy decreases, but $\bar{\sigma}_i$ remains bounded, preventing excessive dominance owing to entropy saturation and cross-defense normalization. Conversely, when all defenses yield similar scores, the weights $\bar{\sigma}_i$ converge and the weights naturally become uniform, reflecting genuine ambiguity rather than harmful dilution. Together with the historical smoothing in Eq. (5), these properties ensure that weight allocation remains moderate and robust even in extreme score configurations.

C. Scoring Defense Strategy Design

Based on the four defense principles and the EWM-based integration, we design four scoring defense strategies.

1) Dictatorship-Based Defense: Honest clients generate "promising" updates by minimizing empirical loss, guiding the global model toward optimal parameters. Malicious clients, as our threat model, aim to degrade accuracy or inject backdoors. Updates from honest clients are inherently more promising. To meet the security requirement R5, we bootstrap a reference model update $\boldsymbol{g}_R^{t+1}$ from client updates, treating it as a promising local update. This approach removes the need for trusted server-side datasets. We use cosine similarity to measure the directional alignment between local and reference model updates, serving as a trust metric for the client. To

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mitigate the impact of negative similarity, negative values are clipped by ReLU function. The trust score is computed as:

$$\boldsymbol{c}_{1,i}{}^{t} = \mathrm{ReLU}\left(\frac{\langle \boldsymbol{g}_R^t, \boldsymbol{g}_i^t\rangle}{\|\boldsymbol{g}_R^t\| \cdot \|\boldsymbol{g}_i^t\|}\right), \tag{6}$$

where $\mathrm{ReLU}(x)=\max\{0, x\}$, $\langle \boldsymbol{g}_R^t, \boldsymbol{g}_i^t\rangle$ inner product of $\boldsymbol{g}_R^t$ and $\boldsymbol{g}_i^t$, and $\frac{\langle \boldsymbol{g}_R^t, \boldsymbol{g}_i^t\rangle}{\|\boldsymbol{g}_R^t\|\cdot\|\boldsymbol{g}_i^t\|}$ is the cosine similarity. If the angle between $\boldsymbol{g}_R^t$ and $\boldsymbol{g}_i^t$ exceeds 90°, $\boldsymbol{c}_{1,i}{}^{t} = 0$.

2) Diversity-Based Defense: Diversity principle effectively mitigates Sybil-based attacks, where malicious clients act as clones with similar updates to drive the global model toward an attack goal. Diversity is key to distinguishing honest clients from Sybil clones, as the latter produce highly similar updates. To accurately calculate diversity, we incorporate historical model updates. Client i's historical model update is: $H_i = \sum_{t=1}^{T} \boldsymbol{g}_i^t$, where T is current global training iteration. S_T-weighted cosine similarity is: $cs_{i,j} = S_T \frac{\langle H_i, H_j\rangle}{\|H_i\|\cdot\|H_j\|}$, where S_T is a weights varying with the iteration. The diversity-based trust score is computed as follows:

$$\boldsymbol{c}_{2,i}{}^{t} = \kappa\left(\ln\left(\frac{a_i}{1-a_i}\right) + 0.5\right) \tag{7}$$

where $a_i = \frac{1-\max_{j\in\mathbb{N}} cs_{i,j}}{\max_{i\in\mathbb{N}}(1-\max_{j\in\mathbb{N}} cs_{i,j})}$ and κ is confidence parameter. Eq. (7) is inspired by FoolsGold [28]. Given the non-IID nature of client data, honest clients are expected to generate diverse updates. The logit transformation sharpens similarity distinctions, penalizing overly similar clients and preserving the influence of those with distinct behaviors.

3) Centroid-Based Defense: When $m < \frac{n}{2}$, clustering methods are effective. At the tth global iteration, we use K-means clustering to split $\{\boldsymbol{g}_i^t\}_{i\in\mathbb{S}^t}$ into two groups G' and G'' with centroids g' and g'', respectively. The cosine similarity $\mathrm{Sim}(x, y) = \frac{\langle x, y\rangle}{\|x\|\cdot\|y\|}$ is used to compute the trust score as:

$$\boldsymbol{c}_{3,i}{}^{t} = \mathrm{ReLU}\left(\frac{\langle \boldsymbol{g}_{\text{cen}}^t, \boldsymbol{g}_i^t\rangle}{\|\boldsymbol{g}_{\text{cen}}^t\| \cdot \|\boldsymbol{g}_i^t\|}\right)$$
$$\text{s.t. } \boldsymbol{g}_{\text{cen}}^t = \begin{cases} g', d_3 > \max\{d_1, d_2\} \wedge |G'| > |G''| \\ g'', d_3 > \max\{d_1, d_2\} \wedge |G'| \le |G''| \\ g', d_3 \le \max\{d_1, d_2\} \wedge d_1 > d_2 \\ g'', d_3 \le \max\{d_1, d_2\} \wedge d_1 \le d_2 \end{cases}, \tag{8}$$

where $d_1 = \mathrm{Sim}(g', \boldsymbol{g}_R^t)$, $d_2 = \mathrm{Sim}(g'', \boldsymbol{g}_R^t)$, $d_3 = \mathrm{Sim}(g', g'')$. A high similarity between g' and g'' may indicate that malicious clients are imitating honest behavior, in which case selecting the cluster center remains a reasonable choice, as it reflects the majority trend. In contrast, a low similarity suggests that malicious clients are significantly deviating from honest updates, making the reference update a more trustworthy and safer option for aggregation.

4) Trigger Detection-Based Defense: Trigger Detection targets 3DFed [17], a SOTA backdoor attack that manipulates specific neurons (indicators) to monitor whether the server accepts a client's update. Honest clients rarely alter these neurons, whereas malicious clients must modify them to sustain the attack. Thus, we design an indicator, i.e., a set of redundant neurons with minimal impact on the main task. Any client modifying these neurons is flagged as suspicious and likely malicious. Specifically, we identify neurons where model updates have significantly smaller magnitudes compared to others. Let $\boldsymbol{g}_G^t(I)$ be the parameters of $\boldsymbol{g}_G^t$ at index I.

$$\hat{I} = \arg\min_{\hat{I}_i} |\boldsymbol{g}_G^t(\hat{I}_i)| \tag{9}$$

where $\hat{I} = \{\hat{I}_1, \hat{I}_2, \ldots, \hat{I}_j\}$ is the index of the set of neurons with the smallest absolute values.

We identify neurons whose parameters minimally impact task loss. Using the Hessian H_G (second derivative of the loss function), the changing direction of model updates is:

$$H_G = \left|\frac{\partial^2 \mathbb{E}_{X\in D_R}[L(X, \boldsymbol{w}^t{}_G)]}{\partial^2 \boldsymbol{w}^t{}_G}\right|, \text{ s.t. } R = \arg\max_{i\in\mathbb{N}} \hat{q}_i^{t-1}. \tag{10}$$

The indicator is computed with $\bar{I} = \arg\min_{\bar{I}_i\in\hat{I}} |H_G(\bar{I}_i)|$, where $H_G(\mathrm{I})$ denotes the parameter in H_G at index I. Redundant neurons depend on the network structure and dataset, remaining constant across iterations. Thus, the index $\bar{I}$ can be reused after calculation. At t^{th} global training iteration, the server receives local model updates $\{\boldsymbol{g}_i^t : i \in \mathbb{S}^t\}$, and extracts values at index $\bar{I}$, i.e., $\{\boldsymbol{g}_i^t(\bar{I}) : i \in \mathbb{S}^t\}$. We compute the required Hessian–vector products once at initialization, restricted to the last layer and a small trusted-client batch, resulting in a one-time $O(nd)$ overhead, where d is the dimension of the model updates and n is the number of clients.

We use the iForest algorithm [49], [50] to detect outliers in the $\{\boldsymbol{g}_i^t(\bar{I}) : i \in \mathbb{S}^t\}$ set. iForest is an unsupervised algorithm that isolates data points via recursive random partitioning, where outliers are separated in fewer splits than normal points. The steps are as follows: (1) Split $\{\boldsymbol{g}_i^t(\bar{I}) : i \in \mathbb{S}^t\}$ into 5 subsets (5)-fold cross-validation); (2) Initialize the iForest (referred to as the IsF for short) and define parameter values controlling its sensitivity and complexity; (3) During training, the IsF builds decision trees based on the subset of $\{\boldsymbol{g}_i^t(\bar{I}) : i \in \mathbb{S}^t\}$ and evaluates if points in the validation set are outliers; (4) Retrain the IsF with optimal parameters and detect outliers in the entire dataset, generating a prediction for each point. Let the iForest be $\mathrm{IsF}(\{\boldsymbol{g}_i^t(\bar{I}) : i \in \mathbb{S}^t\}, i) = \{1, -1\}$, where 1 indicates that $\boldsymbol{g}_i^t(\bar{I})$ is normal and -1 indicates that $\boldsymbol{g}_i^t(\bar{I})$ is an outlier. The trust score is:

$$\boldsymbol{c}_{4,i}{}^{t} = \begin{cases} 1, & \mathrm{IsF}(\{\boldsymbol{g}_i^t(\bar{I}) : i \in \mathbb{S}^t\}, i) = 1 \\ 0, & \mathrm{IsF}(\{\boldsymbol{g}_i^t(\bar{I}) : i \in \mathbb{S}^t\}, i) = -1 \end{cases}. \tag{11}$$

D. EndPCA Algorithm

Algorithm 1 presents EndPCA for the server-side. It run s T global iterations. In each round, the server distributes the current global model to the selected clients, collects their local updates, and then computes the reference update based on these uploads (lines 6 to 10). Next, the server computes the trust scores using the four scoring defense strategies and computes the importance weights $\bar{\sigma}_j$ for all strategies (lines 11 to 16). In line 12, DIC, DIV, CEN, and TRD represent dictatorship-based defense, diversity-based defense, centroid-based defense and trigger detection-based defense, respectively. In lines 17 to 25, the server computes the ensemble trust scores $\hat{q}_i^t$ for all clients. Finally, the server updates the global model using the aggregated model updates (line 26 and 27). Fig. 3 illustrates the operations and algorithms involved in update aggregation on the server during each global training iteration (corresponding to lines 8 to 26 in Algorithm 1). To calculate trust scores in DIV, CEN, and TRD, EndPCA invokes FoolsGold,[1] the K-means algorithm,[2] and 3DFed.[3] Our code is open-source.[4] As for computation overhead, DIC computes inner products per client, with cost $O(kd)$,

[1]https://github.com/DistributedML/FoolsGold
[2]https://scikit-learn.org/stable/api/sklearn.cluster.html
[3]https://github.com/haoyangliASTAPLE/3DFed
[4]https://github.com/GeorgeDUT/fedlearning

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Algorithm 1 EndPCA for Server-Side

Input: Global learning rate α, number of training iterations T;
Output: Global model parameter $\boldsymbol{w}^T{}_G$.
Randomly initialize $\boldsymbol{w}^0{}_G$, and initialize $\boldsymbol{g}_R^0$ with 0;
Initialize each $\hat{q}_i^0$ with 0, where $i \in \{1, \ldots, |\mathbb{N}|\}$;
for $t = 1$ **to** T **do**
  Sample a client set $\mathbb{S}^t$ from $\mathbb{N}$ and send $\boldsymbol{w}^t{}_G$ to them;
  Wait for all clients in $\mathbb{S}^t$ to finish the local training;
  Receive $\{\boldsymbol{g}_i^t : i \in \mathbb{S}^t\}$ from the selected clients;
  $k \leftarrow \arg\max_{i \in \mathbb{S}^t} \hat{q}_i^{t-1}$;
  $\boldsymbol{g}_R^t \leftarrow (\boldsymbol{g}_k^t + (t-1)\boldsymbol{g}_R^{t-1})/t$;
  for $i \in \mathbb{S}^t$ **do**
    Compute $\boldsymbol{c}_{1,i}{}^t, \boldsymbol{c}_{2,i}{}^t, \boldsymbol{c}_{3,i}{}^t$, and $\boldsymbol{c}_{4,i}{}^t$ using DIC, DIV, CEN, and TRD, respectively;
  end for
  for each defense strategy i **do**
    Compute $\bar{\sigma}_i$ of $\{\boldsymbol{c}_{i,j} : j \in \mathbb{S}^t\}$ using Eq. (2);
  end for
  for $i \in \mathbb{N}$ **do**
    if $i \in \mathbb{S}^t$ **then**
      $q_i^t \leftarrow \sum_{j=1}^4 \bar{\sigma}_j \boldsymbol{c}_{j,i}$; // There are 4 defense strategies.
      $\alpha_q \leftarrow \max\{1, 2|q_i^t - \hat{q}_i^{t-1}| - (q_i^t - \hat{q}_i^{t-1})^2\}$;
      $\hat{q}_i^t \leftarrow \hat{q}_i^{t-1} + \alpha_q(q_i^t - \hat{q}_i^{t-1})$;
    else
      $\hat{q}_i^t \leftarrow \hat{q}_i^{t-1}$;
    end if
  end for
  $\boldsymbol{g}_G^t \leftarrow \frac{1}{\sum_{j \in \mathbb{S}^t} \hat{q}_j^t} \sum_{i \in \mathbb{S}^t} \hat{q}_i^t \frac{\|\boldsymbol{g}_R^t\|}{\|\boldsymbol{g}_i^t\|} \boldsymbol{g}_i^t$;
  $\boldsymbol{w}^t{}_G \leftarrow \boldsymbol{w}^{t-1}{}_G - \alpha \boldsymbol{g}_G^t$;
end for

where d is the dimension of the model updates, k is the number selected clients. DIV evaluates similarities across client pairs, yielding $O(dn^2)$. CEN performs K-means and similarity with complexity $O(nd)$. TRD requires Hessian–vector products with complexity $O(nd)$. Among these modules, the DIV is the most sensitive to increases in the number of participating clients, as its pairwise similarity computation scales as $O(dn^2)$. As for communication overhead, EndPCA exchanges only standard model updates, identical in format to FedAvg. Each round incurs nd downlink and kd uplink cost (with $k \ll n$), so the communication overhead remains on par with existing robust aggregation methods. These components are selected to collectively satisfy the defined security requirements. As shown in Section V-E, removing any one of them leads to degraded performance, confirming their necessity in EndPCA.

E. Formal Analysis

To analyze the convergence of EndPCA, we introduce the following three assumptions (the gap between a random variable's expectation and sample mean, the loss function, and the gradient upper bound).

Assumption 1: If $X_1, \ldots, X_n$ are independent random variables where X_i's are sub-exponential with scaling parameters (ν_i, α_i) and mean μ_i, and $X_1', \ldots, X_n'$ are independent random variables that are obtained by sampling from the probability distribution of X_i according to a sampling policy π, then $\sum_{i=1}^n X_i'$ is sub-exponential scaling parameters (ν_*, α_*), where $\nu_*^2 = \sum_{i=1}^n \nu_i^2$ and $\alpha_* = \max_{1 \le i \le n} \alpha_i$. Moreover,

$$\mathbb{P}\left\{\sum_{i=1}^n (X_i' - \mu_i) \ge (t + \mathcal{B}(\pi))\right\} \le \begin{cases} \exp(-t^2/(2\nu_*^2)), & \text{if } 0 \le t \le \nu_*^2/\alpha_* \\ \exp(-t/(2\alpha_*)), & \text{otherwise} \end{cases}, \tag{12}$$

where $\mathcal{B}(\pi)$ denotes the bounded bias induced by π. □

Note that Assumption 1 introduces an additional sampling policy π based on Theorem C.2 from [51]. π will introduce a bounded bias, and the bias will ultimately affect the upper bound of the difference between $\boldsymbol{w}^*$ and $\boldsymbol{w}^t{}_G$. In EndPCA, the server does not rely on a clean and unbiased root dataset to form a reference update. Instead, it constructs this reference using the updates from a subset of high-trust clients. Consequently, the server operates on an empirical mean X_i', derived from these trustworthy clients, which may differ slightly from the true local mean X_i. The quantity $\mathcal{B}(\pi)$ captures this deviation introduced by the client-selection policy π.

Assumption 2: The expected loss function $\mathcal{L}(\boldsymbol{w})$ is μ-strongly convex and differentiable over the parameter space Ω with $\mathcal{K}$-Lipschitz gradient. That is, for all $\boldsymbol{w}, \boldsymbol{w}' \in \Omega$,

$$\mathcal{L}(\boldsymbol{w}') \ge \mathcal{L}(\boldsymbol{w}) + \langle \nabla\mathcal{L}(\boldsymbol{w}), \boldsymbol{w}' - \boldsymbol{w}\rangle + \frac{\mu}{2}\|\boldsymbol{w}' - \boldsymbol{w}\|^2,$$
$$\|\nabla\mathcal{L}(\boldsymbol{w}) - \nabla\mathcal{L}(\boldsymbol{w}')\| \le \mathcal{K}\|\boldsymbol{w} - \boldsymbol{w}'\| \tag{13}$$

where $\langle \cdot, \cdot \rangle$ represent inner product of two vectors. In addition, the empirical loss function $L(D, \boldsymbol{w})$ is $\mathcal{K}_1$-Lipschitz probabilistically. That is, for any $\delta \in (0, 1)$, there exists an $\mathcal{K}_1$ such that: $\mathbb{P}\left\{\sup_{\boldsymbol{w}, \boldsymbol{w}' \in \Omega: \boldsymbol{w} \ne \boldsymbol{w}'} \frac{\|\nabla L(D, \boldsymbol{w}) - \nabla L(D, \boldsymbol{w}')\|}{\|\boldsymbol{w} - \boldsymbol{w}'\|} \le \mathcal{K}_1\right\} \ge 1 - \frac{\delta}{3}$. □

Assumption 3: The gradient of the empirical loss function $L(D, \boldsymbol{w}^*)$ at the optimal global model parameters $\boldsymbol{w}^*$ is bounded. For any $\boldsymbol{w} \in \Omega$, the gradient difference $h(D, \boldsymbol{w}) = \nabla L(D, \boldsymbol{w}) - \nabla L(D, \boldsymbol{w}^*)$ is bounded. Specifically, there exist positive constants δ_1 and γ_1 such that for any unit vector $\boldsymbol{v}$, $\langle \nabla L(D, \boldsymbol{w}^*), \boldsymbol{v}\rangle$ is sub-exponential with σ_1 and γ_1. And for any unit vector $\boldsymbol{v}$, and $\forall \boldsymbol{w} \in \Omega$ with $\boldsymbol{w} \ne \boldsymbol{w}^*$, there exist positive constants δ_2 and γ_2 such that $\langle h(D, \boldsymbol{w}) - \mathbb{E}[h(D, \boldsymbol{w})], \boldsymbol{v}\rangle / \|\boldsymbol{w} - \boldsymbol{w}^*\|$ is sub-exponential with σ_2 and γ_2. Formally, for $\forall |\eta| \le \frac{1}{\gamma_1}$, and $\forall |\eta| \le \frac{1}{\gamma_2}$, the following formulas are always true:

$$\sup_{\boldsymbol{v} \in \mathbb{B}} \mathbb{E}\left[\exp(\eta\langle \nabla L(D, \boldsymbol{w}), \boldsymbol{v}\rangle)\right] \le \exp\left(\frac{\sigma_1^2\eta^2}{2}\right),$$
$$\sup_{\boldsymbol{w} \in \Omega, \boldsymbol{v} \in \mathbb{B}} \mathbb{E}\left[\exp\left(\frac{\eta\langle h(D, \boldsymbol{w}) - \mathbb{E}[h(D, \boldsymbol{w})], \boldsymbol{v}\rangle}{\|\boldsymbol{w} - \boldsymbol{w}^*\|}\right)\right] \le \exp\left(\frac{\sigma_2^2\eta^2}{2}\right) \tag{14}$$

where $\mathbb{B} = \{\boldsymbol{v}' : \|\boldsymbol{v}'\| = 1\}$ is a unit sphere. □

Based on the assumptions, we derive two conclusions on EndPCA: (1) as the global training iteration t increases, the difference between $\boldsymbol{w}^*$ and $\boldsymbol{w}^t{}_G$ decreases; (2) as $t \to \infty$, this difference converges to a bounded value. Here, $\boldsymbol{w}^*$ is the global model's optimal parameters on $\mathcal{X}$, and $\boldsymbol{w}^t{}_G$ represents the server's model parameters after the tth training iteration. These two conclusions can be formally stated as Theorem 1.

Theorem 1: Suppose Assumptions 1-3 hold. In addition, assume that $\mathcal{B}(\pi) \le \Delta_2$. For an arbitrary number of malicious clients but at least one honest client, the difference between

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the global model learned by EndPCA and the optimal global model $\boldsymbol{w}^*$ is bounded. We have the following inequality with probability at least $1-\delta$:

$$\|\boldsymbol{w}^t{}_G - \boldsymbol{w}^*\| \leq (1-\rho)^t \|\boldsymbol{w}^0{}_G - \boldsymbol{w}^*\| + \rho^{-1}\alpha\left(2+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\right)(5\Delta_1+2\Delta_2), \tag{15}$$

where $\boldsymbol{w}^t{}_G$ denotes the parameters of global model in the tth iteration, $\rho = 1 - \left(\sqrt{1-\frac{\mu^2}{4\mathcal{K}^2}} + 12\Delta_2\alpha\left(2+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\right) + \alpha\left(1+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\right)\mathcal{K}\right)$, α is the global learning rate, $\Delta_1 = \sigma_1\sqrt{\frac{2(d\log 6+\log(3/\delta))}{|D_R{}^t|}}$, $\Delta_2 = \sigma_2\sqrt{\frac{2}{|D_R{}^t|}}\sqrt{K_1+K_2}$, $K_1 = d\log\frac{18\mathcal{K}_2}{\sigma_2}$, $K_2 = \frac{d}{2}\log\frac{|D_R{}^t|}{d} + \log\left(\frac{6\sigma_2^2 r\sqrt{|D_R{}^t|}}{\gamma_2\sigma_1\delta}\right)$, $\mathcal{K}_2 = \max\{\mathcal{K},\mathcal{K}_1\}$, $\mathcal{K}$ and $\mathcal{K}_1$ are Lipschitz parameters given in Assumption 2, σ_1 and σ_2 are given in Assumption 3, $D_R{}^t$ is the joint training dataset of all clients in $\mathbb{S}_R^t$, i.e., $D_R{}^t = \bigcup_{i\in\mathbb{S}_R^t} D_i$, $\mathbb{S}_R^t$ consists of the clients corresponding to the maximum $\hat{q}_i$ in each iteration until global training iteration t, $|D_R{}^t|$ is the size of $D_R{}^t$, d is the dimension of model parameters $\boldsymbol{w}$, r is some positive number such that $\|\boldsymbol{w}-\boldsymbol{w}^*\| \leq r\sqrt{d}$ for all $\boldsymbol{w}\in\Omega$. The inequality for r means that the parameter space Ω is constrained. When $|1-\rho|<1$, we have $\lim_{t\to+\infty}\|\boldsymbol{w}^t{}_G - \boldsymbol{w}^*\| \leq \frac{\alpha(2+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t)(5\Delta_1+2\Delta_2)}{\rho}$. □

Theorem 1 shows that EndPCA can converge to a bounded neighborhood around the optimum without relying on a trusted dataset. According to Eq. (15), as $|D_R{}^t|$ increases, the difference between $\boldsymbol{w}^*$ and $\boldsymbol{w}^t{}_G$ decreases. Given a fixed D_i, we can enlarge $D_R{}^t$ by increasing $\mathbb{S}_R^t$. However, clients added to $\mathbb{S}_R^t$ must be honest, so we cannot include all clients in $\mathbb{S}_R^t$. Therefore, at each global training iteration, we select the local model update corresponding to the largest $\hat{q}_i$ value and compute a weighted average with the historical model updates. This approach expands $\mathbb{S}_R^t$ while ensuring a high probability that clients in $\mathbb{S}_R^t$ are honest. $\mathcal{B}(\pi)$ enlarges the radius of the bounded region around the optimum $\boldsymbol{w}^*$. But it does not affect the descent dynamics or the existence of convergence. $\mathcal{B}(\pi) \leq \Delta_2$ appears only as an additive term in the bound, reflecting the statistical mismatch between the server's trusted estimate and the true underlying distribution.

To establish Theorem 1, we decompose its proof into four supporting lemmas (i.e., Lemma 1 to Lemma 4). Each lemma is proved individually, and the theorem is subsequently derived based on the results of these lemmas. The core idea of the proof is as follows: directly proving the upper bound of $\|\boldsymbol{w}^t{}_G - \boldsymbol{w}^*\|$ is challenging, so we use $\boldsymbol{g}_G^t$, $\nabla\mathcal{L}()$, $\nabla L(,)$, etc., as intermediate bridges and respectively prove the relationships among them. Then, by the triangle inequality and ϵ-net argument theory, we establish the relationship between $\boldsymbol{w}^*$ and $\boldsymbol{w}^t{}_G$. The final result is an upper bound of the $\|\boldsymbol{w}^t{}_G - \boldsymbol{w}^*\|$, but not its supremum. In practical experiments, the actual difference is much smaller than this upper bound. The detailed proof steps are provided in the appendix.

Lemma 1: For an arbitrary iteration and an arbitrary number of malicious clients, the distance between global model update $\boldsymbol{g}_G^t$ and the gradients of the expectation loss $\nabla\mathcal{L}(\boldsymbol{w}^t{}_G)$ is bounded as follows:

$$\|\boldsymbol{g}_G^t - \nabla\mathcal{L}(\boldsymbol{w}^t{}_G)\| \leq \left(2+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\right)\|\boldsymbol{g}_R^t - \nabla\mathcal{L}(\boldsymbol{w}^t{}_G)\| + \left(1+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\right)\|\nabla\mathcal{L}(\boldsymbol{w}^t{}_G)\| \tag{16}$$

where $\boldsymbol{g}_G^t = \frac{1}{\sum_{j\in\mathbb{S}^t}\hat{q}_j^t}\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\frac{\|\boldsymbol{g}_R^t\|}{\|\boldsymbol{g}_i^t\|}\boldsymbol{g}_i^t$. $\boldsymbol{g}_R^t$. □

Lemma 2: If we set the learning rate as $\alpha = \frac{\mu}{2\mathcal{K}^2}$, then we have the following in any global training iteration $t \geq 1$:

$$\|\boldsymbol{w}^{t-1}{}_G - \boldsymbol{w}^* - \alpha\nabla\mathcal{L}(\boldsymbol{w}^{t-1}{}_G)\| \leq \sqrt{1-\frac{\mu^2}{4\mathcal{K}^2}}\|\boldsymbol{w}^{t-1}{}_G - \boldsymbol{w}^*\| \tag{17}$$

where μ and $\mathcal{K}$ are defined in Assumption 2. □

Lemma 3: Given any $\delta \in (0,1)$ and $\boldsymbol{w} \in \Omega$. If Assumption 3 holds, $\Delta_1 \leq \sigma_1^2/\gamma_1$ and $\Delta_3 \leq \sigma_2^2/\gamma_2$, then it holds that:

$$\mathbb{P}\left\{\left\|\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla L(X,\boldsymbol{w}^*) - \nabla\mathcal{L}(\boldsymbol{w}^*)\right\| \geq 2\Delta_1 + 2\mathcal{B}(\pi)\right\} \leq \frac{\delta}{3}, \text{ and} \tag{18}$$

$$\mathbb{P}\left\{\left\|\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla h(X,\boldsymbol{w}^*) - \mathbb{E}_{X\sim\mathcal{X}}[h(X,\boldsymbol{w})]\right\| \geq 2\Delta_3\|\boldsymbol{w}-\boldsymbol{w}^*\| + 2\mathcal{B}(\pi)\right\} \leq \frac{\delta}{3} \tag{19}$$

where $\Delta_1 = \sigma_1\sqrt{\frac{2(d\log 6+\log(3/\delta))}{|D_R{}^t|}}$ and $\Delta_3 = \sigma_2\sqrt{\frac{2(d\log 6+\log(3/\delta))}{|D_R{}^t|}}$. □

Lemma 4: Suppose Assumption 1-3 and $\mathcal{B}(\pi) \leq \Delta_2$ hold, and $\Omega \subset \{\boldsymbol{w} : \|\boldsymbol{w}-\boldsymbol{w}^*\| \leq r\sqrt{d}\}$ holds for some positive parameter r, where d is the dimensionality of the global model parameters. Then, given $\forall\delta \in (0,1)$ and $\forall\boldsymbol{w} \in \Omega$, if $\Delta_1 \leq \frac{\sigma_1^2}{\gamma_1}$ and $\Delta_2 \leq \frac{\sigma_2^2}{\gamma_2}$, the following formula always holds:

$$\mathbb{P}\{\|\boldsymbol{g}_R^t - \nabla\mathcal{L}(\boldsymbol{w})\| \leq (12\Delta_2\|\boldsymbol{w}-\boldsymbol{w}^*\| + 5\Delta_1 + 2\Delta_2)\} \geq 1-\delta \tag{20}$$

where $\Delta_2 = \sigma_2\sqrt{\frac{2}{|D_R{}^t|}}\sqrt{K_1+K_2}$, $K_1 = d\log\frac{18\mathcal{K}_2}{\sigma_2}$, $K_2 = \frac{d}{2}\log\frac{|D_R{}^t|}{d} + \log\left(\frac{6\sigma_2^2 r\sqrt{\sum_{i\in\mathbb{S}_R^t}|D_i|}}{\gamma_2\sigma_1\delta}\right)$, $\mathcal{K}_2 = \max\{\mathcal{K},\mathcal{K}_1\}$, and $|D_i|$ is the size of local training dateset on client i. $\mathcal{B}(\pi)$ is the sampling biases defined in Assumption 1. □

V. Evaluation

A. Experimental Setup

1) Comparison BRARs and Attacks: We provide extensive experiments to evaluate the effectiveness of EndPCA, by comparing the accuracy of the learned global model and attack success rate with FedAvg [32] and several comparison BRARs, including FLTrust [26], FLest [42], Foolsgold [28], FLDetector [24], FLAME [29], RFLBAT [39], Deepsight [19], and BackdoorIndicator [40]. We consider several poisoning attacks: Labelflip (LF) [38], Updateflip (UF) [15], Trim [14], Adaptive attack [14], Scaling attack (SA) [16], 3DFed [17], ALIE [52], IPM [53], and DistanceMaximization [54]. The settings for LF, UF, Trim and Adaptive follow that in the FLTrust [26],

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while SA and 3DFed use configurations from their original papers [16], [17]. ALIE, IPM and DistanceMaximization are implemented based on open-source code.[5]

2) Benchmark: Following the experimental setup in the previous work [17], [20], [23], [24], [26], [55], we evaluate our method on the following datasets: MNIST [56], Fashion-MNIST [57], CIFAR10 [58], Tiny-Imagenet [59], and HAR [60]. They are popular in FL community for evaluating the effectiveness of attack and defense methods. These datasets cover a range of dimensions, including lightweight to large-scale datasets, small size to large size, grayscale to color images, and specific object classification to general object classification. The neural network architectures include MLP, CNNs FCNs, and Resnet18. To simulate the IID and non-IID datasets, we employ the label distribution [61]. The implementation aligns with the method described in [17], [26], and [42]. In our implementation, non-IID data are generated using a Dirichlet-based label-partitioning scheme. For each class, we draw a length-n probability vector $(p_1, \ldots, p_n) \sim \mathrm{Dir}(\alpha)$ and assign the class's samples to the n clients proportionally according to these probabilities. This procedure is repeated independently for all classes, yielding a heterogeneous dataset whose class distributions differ across clients. We set the Dirichlet parameter as $\alpha = b$, where b denotes the bias probability: values of b close to 1 produce more uniform (near-IID) allocations, whereas values close to 0 yield highly skewed, strongly non-IID partitions. Other FL settings align with those in the previous works [17], [26].

3) Metrics: We evaluate robustness and fidelity using *model accuracy* (ACC), which is the ratio of correctly classified test inputs. The backdoor resilience is evaluated through *attack success rate* (ASR), which is the ratio of attacker-targeted inputs misclassified as the target label. Metrics are computed with a test batch size of 100. Higher ACC indicates greater robustness, lower ASR reflects stronger backdoor resilience, and fidelity is measured by how closely the ACC matches FedAvg in non-attack scenarios.

Summary of Experimental Results. We highlight the following conclusions regarding security requirements. **Defense-integration robustness**: EndPCA shows the highest robustness and significantly reduces backdoor attack success rates. **Fidelity**: EndPCA achieves model accuracy comparable to FedAvg. **Data distribution independence**: EndPCA maintains robustness and backdoor resilience under both IID and non-IID datasets. **Large-scale malicious clients resilience**: EndPCA consistently delivers strong robustness and backdoor resilience across varying malicious client proportions. Moreover, EndPCA does not rely on any auxiliary trusted dataset, satisfying security requirement R5.

B. Robustness, Resilience and Fidelity

Table II presents the ACC and ASR results. FedAvg attains the highest accuracy in the no-attack setting, as it incurs no defensive overhead, but it fails under adversarial conditions. In the "No attack" row, we therefore assess which BRAR best preserves the clean performance of FedAvg, providing a direct measure of Fidelity. The bias probability is set to 0.9 (refer to Section V-C). We can observe the following: (1) Overall, EndPCA achieves the highest ACC in most cases compared to other BRARs. EndPCA's ACC surpasses the second-best BRAR by 0.12% to 7.22%. (2) In rare cases, EndPCA does not achieve the highest ACC (e.g., under the LF attack on the Tiny-Imagenet dataset). Even in these cases, our method is only 1.03% behind the best-performing method (FLDetector). (3) In all cases, our method achieves the lowest ASR, outperforming the next best BRAR by 0.4% to 74.02%. We can conclude that EndPCA effectively satisfies security requirements R1. Focusing on the "No attack" row in Table II, we observe that: (1) In the absence of attacks, compared to other BRARs, EndPCA's ACC is the closest to FedAvg's ACC on datasets Fashion-MNIST, CIFAR10, and Tiny-Imagenet, with a difference of only 0.22% to 0.38%. (2) Although EndPCA does not achieve the closest ACC to FedAvg on either MNIST or HAR, it ranks second on both datasets, with only a 0.07% gap on MNIST and 0.29% on HAR. These results demonstrate EndPCA satisfies R2.

We also compare our method with the latest defense techniques, BackdoorIndicator [40], specifically designed for 3DFed. BackdoorIndicator provides no convergence guarantee, whereas EndPCA is provably convergent. It focuses solely on backdoor attacks and requires a server-side trusted dataset, while EndPCA covers a broader range of attacks without relying on such data. As shown in Table III, EndPCA outperforms BackdoorIndicator against SA but is slightly less effective against 3DFed.

To demonstrate that the ensemble trust scores can distinguish between honest and malicious clients, Fig. 4 shows the curve of score changes. The x-axis represents the training rounds (starting from the round when the attack begins), and the y-axis represents the normalized ensemble trust score. The results demonstrate that EndPCA effectively distinguishes between honest and malicious clients based on their scores.

C. Impact of IID and Non-IID Datasets

Fig. 5 shows the performance of various methods under different attacks on both IID and non-IID client datasets. The dataset is MNIST. There are 20 clients, of which 4 are malicious ones. The x-axis represents bias probability b (from 0.9 to 0.1), while the y-axis for (a) to (f) is ACC, and for (g) and (h) is ASR. From Fig. 5 (a) to (f), we observe the following: as b decreases, the ACC metric for all methods declines; however, EndPCA's decline is significantly less compared to other methods (except for those whose ACC is already low and has little room to decrease). From Fig. 5 (g) and (h), we observe the following: as b decreases, EndPCA's ASR remains nearly unchanged, whereas the ASR of all other methods increases. Notably, under the SA, although Foolsgold, FLAME, RFLBAT, and Deepsight have ASR metrics close to ours when $b = 0.9$, their ASR metrics rapidly surge to nearly 99% as b drops to 0.7. Similar to Fig. 5, Fig. 6 and Fig. 7 report results on the Fashion-MNIST and HAR datasets. The trends are consistent with those in Fig. 5: our method remains the best (and occasionally second-best) across different bias probabilities. These results demonstrate that though non-IID data affect the performance of EndPCA, it outperforms the comparison methods under various data distributions ranging from IID to non-IID, thereby better satisfying requirement R3.

D. Impact of Proportions of Malicious Clients

Fig. 8 illustrates the performance of various comparison methods under different proportions of malicious clients. The x-axis represents the proportion of malicious clients, while the y-axis for (a) to (f) shows ACC, and for (g) and (h) shows

[5]https://github.com/lishenghui/blades

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TABLE II

THE ACC (%) OF DIFFERENT FL METHODS UNDER DIFFERENT ATTACKS AND THE ASR (%) OF THE 3DFED ATTACK AND SA.THE RESULTS FOR THE 3DFED ATTACK AND SA ARE IN THE FORM OF "ACC(%) / ASR (%)".THE NON-IID SETTINGS ALIGN WITH THOSE IN [17].THE BEST AND SECOND BEST RESULTS AMONG BRARSARE HIGHLIGHTED IN THE BOLD-RED AND BOLD-BLUE

(a) MLP global model, MNIST, 20 clients, 4 malicious clients

Attack	FedAvg	FLTrust	FLest	Foolsgold	FLDetector	FLAME	RFLBAT	Deepsight	EndPCA
No attack	99.24	98.23	97.97	98.10	**99.20**	98.95	98.12	98.01	**99.17**
LF	98.30	98.54	**98.62**	98.35	93.15	98.22	98.34	98.07	**98.74**
Trim	9.74	**98.94**	91.32	98.61	12.93	98.86	98.80	97.91	**98.88**
Adaptive	11.94	10.06	9.80	10.04	95.32	98.15	98.14	**98.89**	**99.13**
UF	9.82	**98.64**	11.35	98.39	98.32	98.45	98.55	97.79	**98.71**
3DFed	98.03/94.10	97.65/**61.90**	99.03/74.91	99.02/68.36	98.52/78.86	98.90/95.40	98.75/94.85	99.01/78.85	99.05/**9.71**
SA	98.83/84.97	94.59/99.99	97.8/99.71	98.18/**10.62**	95.84/88.79	98.23/10.66	98.4/10.67	98.17/13.9	98.18/**10.29**

(b) CNN global model, Fashion-MNIST, 20 clients, 4 malicious clients

Attack	FedAvg	FLTrust	F1est	Foolsgold	FLDtector	FLAME	RFLBAT	Deepsight	EndPCA
No attack	89.20	88.25	86.45	88.14	88.35	87.53	**88.73**	88.01	**88.82**
LF	86.89	87.32	83.00	**87.61**	87.34	87.45	87.49	87.36	**87.58**
Trim	10.03	11.00	75.89	**88.05**	86.06	84.69	86.10	87.77	**89.16**
Adaptive	10.00	85.03	77.06	**87.63**	85.62	85.22	86.88	81.02	**88.47**
UF	86.35	85.90	83.33	86.66	87.59	87.45	87.59	**88.15**	**88.67**
3DFed	89.10/94.00	88.15/87.49	83.17/**85.33**	87.79/86.88	88.06/89.15	87.30/98.39	84.98/99.1	88.93/92.51	88.07/**11.31**
SA	86.69/92.69	88.22/100.0	85.71/99.02	87.51/33.66	77.91/99.61	82.33/10.26	88.39/15.3	87.94/**10.24**	89.15/**9.76**

(c) Resnet global model, CIFAR10, 100 clients, 20 malicious clients

Attack	FedAvg	FLTrust	F1est	Foolsgold	FLDtector	FLAME	RFLBAT	Deepsight	EndPCA
No attack	73.53	51.30	52.74	61.39	62.65	64.75	**71.84**	62.46	**73.31**
LF	51.06	48.64	13.60	50.76	**64.29**	54.98	35.05	32.99	**65.79**
Trim	17.72	32.58	13.08	58.50	**62.22**	57.37	56.91	60.15	**69.44**
Adaptive	14.08	47.54	34.65	51.70	13.04	55.86	**68.65**	60.02	**72.84**
UF	31.03	59.36	10.49	49.74	**64.45**	55.51	35.93	50.63	**64.78**
3DFed	64.76/85.64	51.24/46.46	11.49/72.31	62.90/63.22	32.15/77.86	58.72/71.77	70.01/52.01	62.15/**39.5**	72.03/**11.04**
SA	29.16/92.3	44.24/74.63	50.45/53.44	52.84/36.06	19.92/89.67	58.30/11.9	70.42/**10.58**	52.40/60.08	72.29/**10.18**

(d) Resnet global model, Tiny-Imagenet, 100 clients, 20 malicious clients

Attack	FedAvg	FLTrust	FLest	Foolsgold	FLDetector	FLAME	RFLBAT	Deepsight	EndPCA
No attack	68.56	65.80	64.14	**67.89**	67.02	64.85	66.93	66.55	**68.21**
LF	65.61	65.17	26.54	64.68	**65.78**	62.80	**65.54**	65.13	64.75
Trim	57.42	62.67	62.90	**66.50**	64.12	62.11	64.89	64.13	**66.90**
Adaptive	61.65	60.30	16.90	**64.40**	60.27	61.40	63.67	63.92	**67.61**
UF	64.43	64.98	27.10	**66.61**	63.60	63.57	65.05	65.08	**67.30**
3DFed	40.35/80.20	61.74/77.19	64.07/64.18	64.58/68.43	37.34/76.69	64.33/**55.56**	62.07/63.5	65.26/66.9	66.93/**10.51**
SA	11.71/80.54	38.17/79.86	39.02/**30.2**	28.79/78.8	27.43/80.33	64.82/84.0	62.20/83.5	60.94/76.3	66.59/**10.61**

(e) FCN global model, HAR, 20 clients, 4 malicious clients

Attack	FedAvg	FLTrust	FLest	Foolsgold	FLDetector	FLAME	RFLBAT	Deepsight	EndPCA
No attack	90.61	59.11	76.48	90.08	87.53	89.56	**90.56**	89.12	**90.32**
LF	83.79	41.92	65.59	76.76	86.22	**88.36**	87.18	86.31	**89.55**
Trim	86.73	48.44	73.74	86.52	81.56	**88.90**	86.39	85.21	**89.85**
Adaptive	82.77	52.16	69.45	84.20	86.37	87.10	86.92	**87.12**	**87.56**
UF	84.70	49.83	74.24	17.69	87.09	85.49	**87.95**	85.23	**88.98**
3DFed	90.54/86.27	45.68/92.62	72.64/28.86	71.88/97.21	87.15/37.23	88.16/**17.02**	86.83/39.16	87.89/23.43	90.28/**15.36**
SA	35.06/100.0	75.73/100.0	81.62/100.0	87.12/18.75	80.81/19.89	88.23/**17.23**	89.28/99.38	88.12/17.31	88.69/**16.35**

TABLE III

BACKDOOR ROBUSTNESS. EXPERIMENTAL SETUP: CIFAR10, RESNET18, 100 CLIENTS, 20 MALICIOUS CLIENTS (ACC(%)/ASR(%))

Method	SA	3DFed
BackdoorIndicator [40]	65.80/50.11	64.30/6.70
EndPCA	72.29/9.80	72.03/8.04

ASR. From Fig. 8 (a) to Fig. 8 (f), we can observe the following: (1) As the proportion of malicious clients increases, the ACC of all methods decreases. However, EndPCA's decrease is much smaller, and it consistently maintains higher ACC than all compared methods across all malicious-client ratios. (2) The proportion of malicious clients at which EndPCA's ACC starts to decline is higher than that of other methods. EndPCA's ACC generally does not start to decline until the proportion of malicious clients exceeds 60%, whereas other methods start to decline well before 60%. When the proportion of malicious clients is less than 60%, EndPCA's ACC is close to 100%. From Fig. 8 (g) and Fig. 8 (h), we can see:

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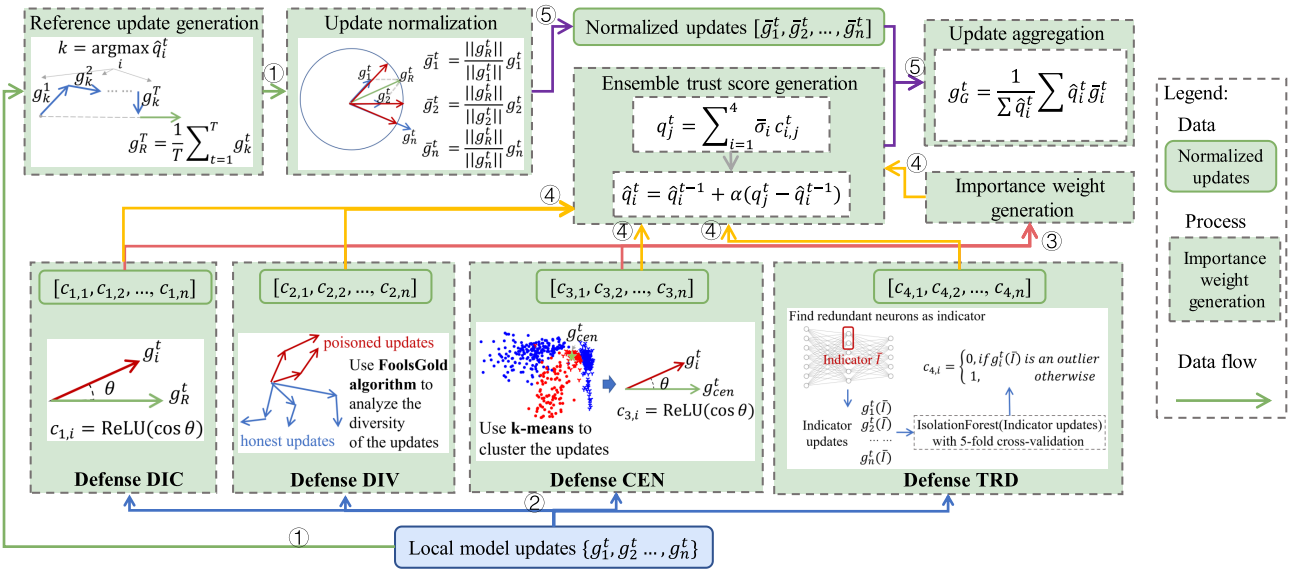
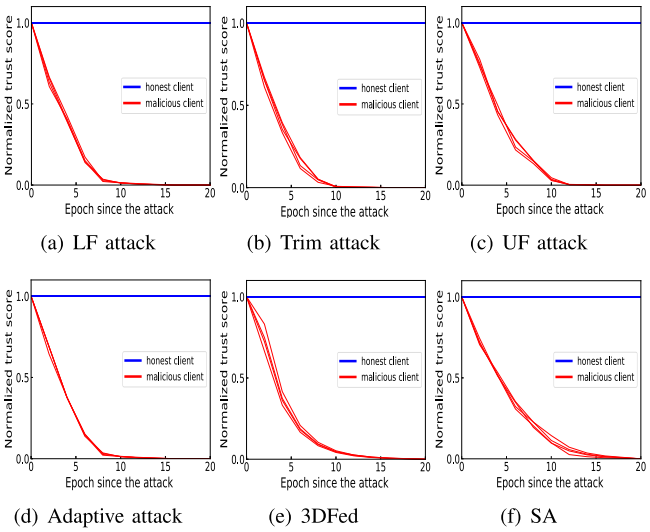


Fig. 3. Illustration of our aggregation function (The numbers represent the sequential steps of each global training iteration in FL).



(a) LF attack (b) Trim attack (c) UF attack

(d) Adaptive attack (e) 3DFed (f) SA

Fig. 4. The dyamics of the normalizated ensemble trust scores of honest clients and malicious clients under EndPCA (MNIST dataset, 20 client, and 10 malicious clients).

As the proportion of malicious clients increases, the ASR for all methods increases to varying degrees. However, our method's ASR remains at a low level of 15% and does not increase further, whereas the ASR for other methods can rise to nearly 100%. These results indicate that while the proportion of malicious clients affects the performance of our method, it still maintains the best robustness and backdoor resilience compared to other methods across different proportions of malicious clients, thus satisfying requirement R4.

E. Ablation Study

To investigate the contribution of different components of EndPCA to its performance, we conduct an ablation study. EndPCA consists of five key modules: the four scoring strategies DIC, DIV, CEN, and TRD, and the history-weighted sum score (i.e., Eq.(3)) in the score-based defense integration

TABLE IV

ABLATION STUDY. RESULTS FOR 3DFED AND SA ARE SHOWN AS "ACC(%) / ASR (%)". THE RESULTS FOR THE OTHER ATTACKS ARE IN THE FORM OF "ACC(%)". W/O X INDICATES THE ABSENCE OF X IN ENDPCA

	No attack	LF	Trim	UF	Adaptive	3DFed	SA
w/o DIC	98.70	98.07	97.74	98.37	95.18	98.14/11.69	98.07/9.61
w/o DIV	99.02	98.19	60.62	96.95	97.03	96.90/10.01	93.82/98.7
w/o CEN	95.16	90.56	90.26	95.01	94.16	93.65/9.7	92.66/9.36
w/o TRD	99.05	98.14	98.32	98.41	98.21	98.12/93.0	98.18/10.29
w/o HWS	99.06	98.74	97.96	97.2	96.09	98.38/9.71	94.85/10.55
EndPCA-H	99.09	98.81	98.84	99.00	98.75	98.08/9.70	98.92/9.66
EndPCA-A	99.10	94.45	79.62	96.65	96.42	95.63/88.17	96.73/93.42
EndPCA	99.17	98.97	98.88	99.03	98.53	98.98/9.65	98.69/9.59

(referred to as HWS below). In particular, EndPCA and five variants of EndPCA are considered. The five variants are: w/o DIC, w/o DIV, w/o CEN, and w/o TRD, which are modified from EndPCA by removing the DIC DIV, CEN, and TRD scoring strategies, respectively, and w/o HWS, which are modified from EndPCA by setting the α_q (in Eq.(3)) to 1. Since DIV uses K-means and HDBSCAN is considered an effective method for clustering model updates [29], [62], we replaced the K-means with HDBSCAN, resulting in the variant EndPCA-H. We also replaced the ensemble trust score generation method with averaging (i.e., averaging the four normalized scores) to create the variant EndPCA-A. Table IV shows the results of the ablation study. The dataset is MNIST. There are 16 honest clients and 4 malicious clients. The bias probability is set to 0.9. On all six poisoning attacks: (1) Removing DIC results in a slight decrease in ACC, particularly under adaptive attack. (2) Removing DIV also leads to a decrease in ACC and a significant increase in ASR. This suggests that the inclusion of DIV plays a crucial role in enhancing our method's resilience to backdoor attacks. (3) Removing CEN leads to a substantial drop in ACC, highlighting its important role in maintaining robustness. (4) Removing TRD significantly increases ASR under the 3DFed attack, demonstrating its specific effectiveness against this attack. (5) Removing HWS leads to a slight decrease in ACC, which shows that HWS

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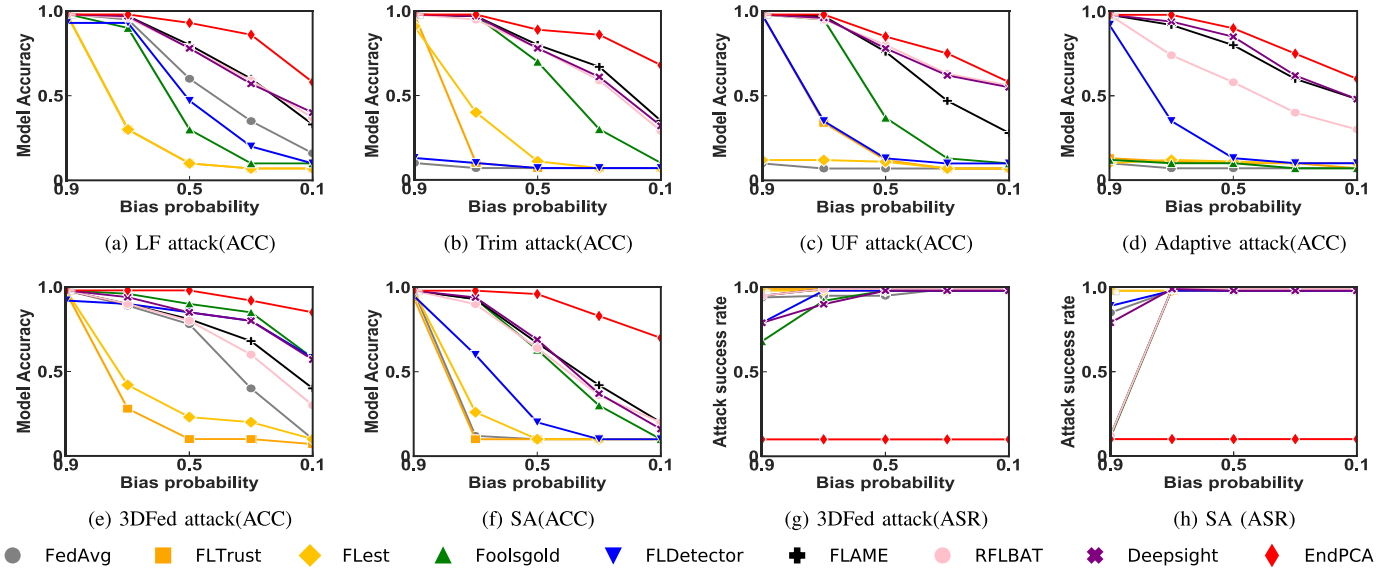
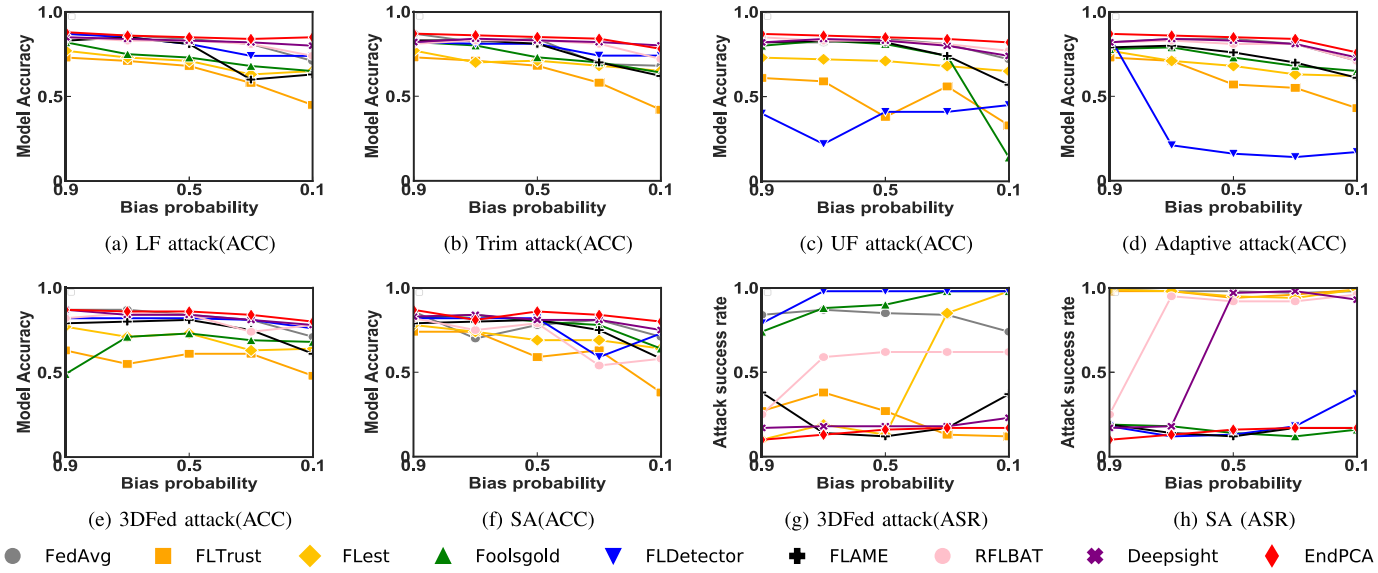


(a) LF attack(ACC) (b) Trim attack(ACC) (c) UF attack(ACC) (d) Adaptive attack(ACC)

(e) 3DFed attack(ACC) (f) SA(ACC) (g) 3DFed attack(ASR) (h) SA (ASR)

Fig. 5. Impact of the degree of non-IID (i.e., bias probability) on ACC/ASR of different methods under different attacks(MNIST).



(a) LF attack(ACC) (b) Trim attack(ACC) (c) UF attack(ACC) (d) Adaptive attack(ACC)

(e) 3DFed attack(ACC) (f) SA(ACC) (g) 3DFed attack(ASR) (h) SA (ASR)

Fig. 6. Impact of the degree of non-IID (i.e., bias probability) on ACC/ASR of different methods under different attacks (Fashion-MNIST).

enhances our method's robustness. (6) EndPCA-H performs similarly to EndPCA under various attacks, indicating that the choice of clustering algorithm has little impact on the overall effectiveness. (7) EndPCA-A shows mediocre performance, significantly worse than EndPCA, highlighting that simple averaging of trust scores reduces the defense's effectiveness.

A key insight of EndPCA is that effective scoring methods tend to produce lower entropy, indicating more confident discrimination of clients. To validate this, we evaluated several methods on MNIST using an MLP model with 20 clients (including 4 malicious ones). For each method, we computed the average entropy over 30 randomly selected rounds. As shown in Table V, and compared with robustness results in Table II (a), methods with lower entropy generally exhibit stronger robustness, supporting our hypothesis.

TABLE V

ENTROPY COMPARISON OF DIFFERENT METHODS

Attack	FedAvg	FLTrust	FLest	Foolsgold	EndPCA
Trim	1.00	0.72	0.78	0.81	0.87
Adaptive	1.00	0.99	0.98	0.98	0.85
UF	1.00	0.92	0.99	0.87	0.84
3DFed	1.00	0.93	0.97	0.97	0.79
SA	1.00	0.99	0.99	0.87	0.80

F. Supplementary Evaluation Under Advanced Attacks

We also conduct supplementary evaluations against several advanced poisoning attacks. The results are summarized in Table VI. The dataset is Fashion-MNIST, and the the neural network model is CNN. There are 16 honest clients and 4 malicious clients. ALIE [52] is a general-purpose poisoning attack framework capable of instantiating both untargeted and targeted objectives. We evaluate two variants: ALIE (Converge), which prevents the model from converging to the optimal solution, and ALIE (Backdoor), which inserts backdoors into the global model. EndPCA maintains strong

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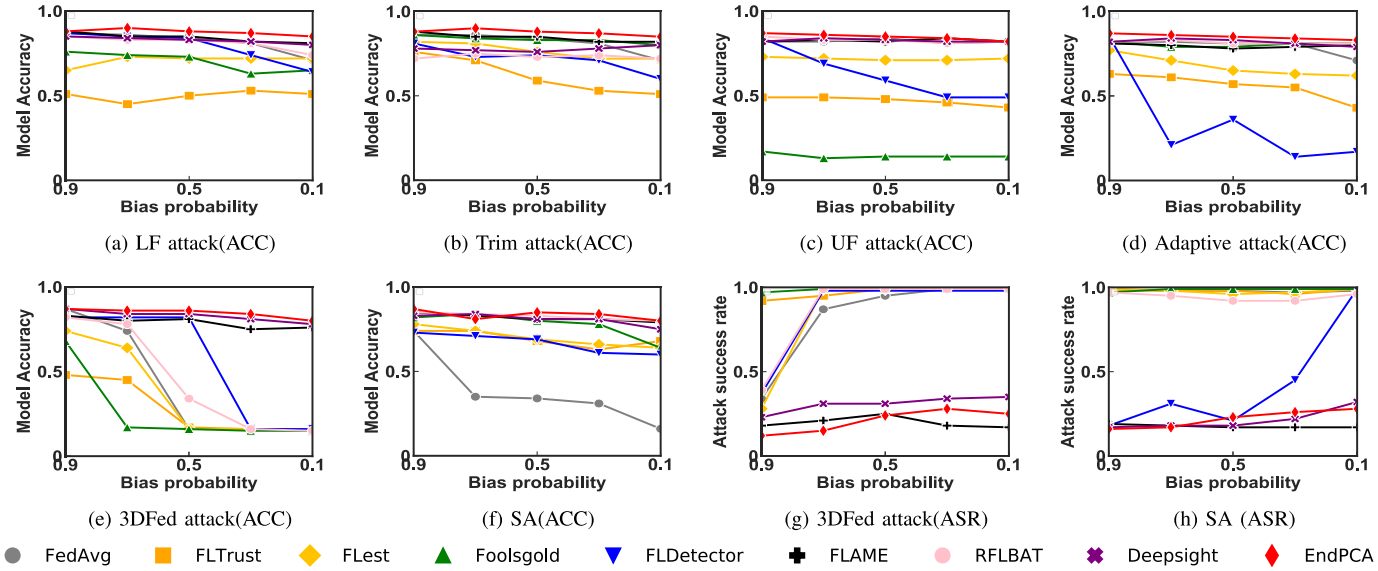
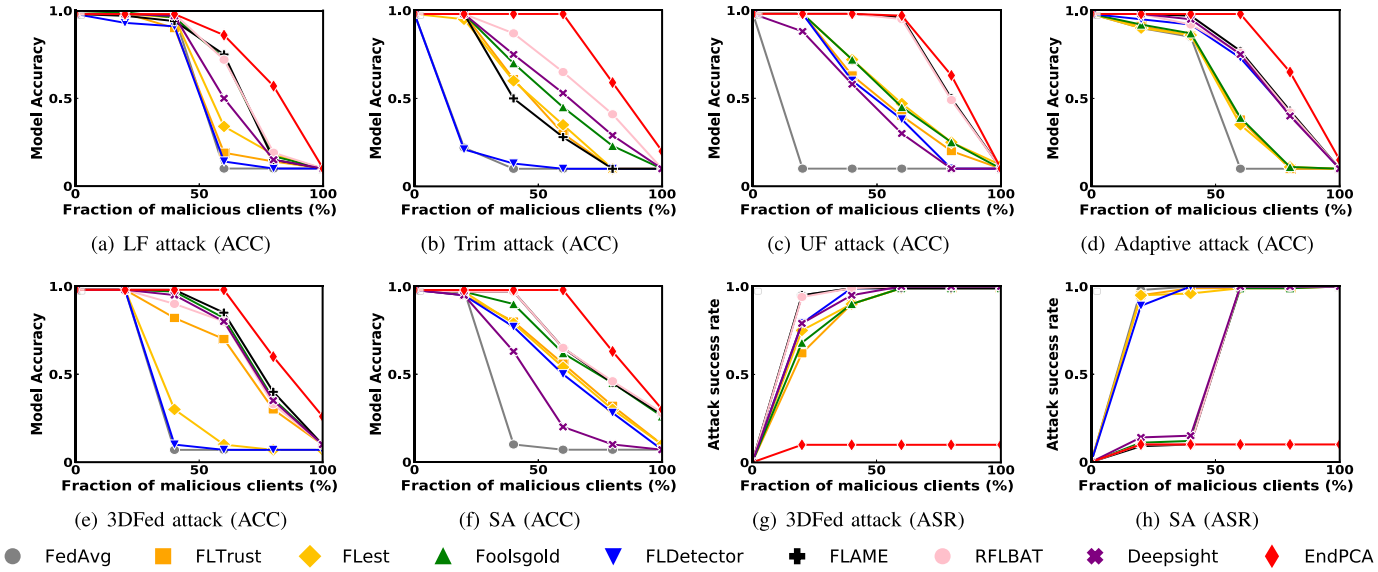


(a) LF attack(ACC) (b) Trim attack(ACC) (c) UF attack(ACC) (d) Adaptive attack(ACC)

(e) 3DFed attack(ACC) (f) SA(ACC) (g) 3DFed attack(ASR) (h) SA (ASR)

Fig. 7. Impact of the degree of non-IID (i.e., bias probability) on ACC/ASR of different methods under different attacks (HAR).



(a) LF attack (ACC) (b) Trim attack (ACC) (c) UF attack (ACC) (d) Adaptive attack (ACC)

(e) 3DFed attack (ACC) (f) SA (ACC) (g) 3DFed attack (ASR) (h) SA (ASR)

Fig. 8. Impact of the fraction of malicious clients on ACC/ASR of different methods under different attacks.

TABLE VI
PERFORMANCE OF ENDPCA UNDER STRONGER ATTACKS

Attack	EndPCA ACC (%) / ASR (%)
ALIE (Converge)	87.98
ALIE (Backdoor)	85.07 / 12.76
IPM	88.59
DistanceMaximization	87.65

performance against both variants, achieving 87.98% accuracy under ALIE (Converge), and 85.07% accuracy with only 12.76% ASR under ALIE (Backdoor). We also test EndPCA against IPM [53] and DistanceMaximization [54]. EndPCA achieves 88.59% and 87.65% ACC under these attacks, respectively, demonstrating its broad-spectrum defense capability even beyond the main evaluation set. We further note that EndPCA is resilient to adaptive adversaries targeting the EWM-based integration. Manipulating the entropy of a single defense requires altering relative trust-score patterns, which either reveals the attacker or leaves the entropy essentially unchanged, while the ensemble still requires evading all heterogeneous defenses simultaneously. Moreover, our bounded-error convergence guarantee ensures stability as long as trust scores remain bounded, even under such adaptive attempts.

VI. CONCLUSION AND DISCUSSION

We developed EndPCA, a broad-spectrum defense against diverse poisoning attacks. Our theoretical analysis proves convergence with bounded error, and experiments show superior performance across datasets, neural networks, and attack settings. This work demonstrates that integrating heterogeneous weak BRARs through an entropy-based dynamic weighting scheme can achieve not only enhanced empirical robustness

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against diverse poisoning attacks, but also provable convergence under mild assumptions without requiring a trusted dataset. This addresses a fundamental limitation of prior defenses by unifying practical effectiveness with theoretical guarantees in federated learning.

There remain several natural avenues for extension. While EndPCA integrates BRARs effectively, it has yet to incorporate other defenses, such as knowledge distillation [63], adversarial training [64], and moving target defense [65]. As for applicability, future work may extend EndPCA to smart-meter networks, ISAC sensing systems, and cyber-physical power infrastructures, where adversarial behaviors may be multimodal, coordinated, or trigger-free [5], [8], [9], [10]. These settings introduce new threat patterns, and adapting the modular scoring framework of EndPCA to such environments is a natural direction for future research. Our theoretical guarantees rely on strong convexity and Lipschitz smoothness to enable a tractable analysis. Extending the convergence analysis of EndPCA to weaker or non-smooth conditions, particularly those reflecting the non-convexity of deep neural networks in practical FL, is an important direction for future work. Such extensions would further enhance the relevance of our theoretical insights to real-world deployments. Another promising direction is to reduce the computational cost of Hessian-based trigger detection, for example by using lightweight curvature approximations, Hessian-free techniques, or alternative trigger-detection mechanisms [66] that avoid explicit second-order computations. Another avenue is to analyze entropy-weighting under adaptive adversaries through a game-theoretic lens, which may help characterize bounded manipulation and further strengthen robustness guarantees.

Appendix
Supplementary Proof of Theorem 1

Now, we begin the proof of Theorem 1. First, by triangle inequality, the left-hand side of inequality (15) in Theorem 1 can be expanded as follows. For any t^{th} global training iteration, we have:

$$\begin{aligned}
&\|\boldsymbol{w}^t{}_G - \boldsymbol{w}^*\| = \|\boldsymbol{w}^{t-1}{}_G - \alpha \boldsymbol{g}_G^{t-1} - \boldsymbol{w}^*\| \\
&= \|\boldsymbol{w}^{t-1}{}_G - \alpha \nabla\mathcal{L}(\boldsymbol{w}^{t-1}{}_G) - \boldsymbol{w}^* + \alpha\nabla\mathcal{L}(\boldsymbol{w}^{t-1}{}_G) - \alpha \boldsymbol{g}_G^{t-1}\| \\
&\le \|\boldsymbol{w}^{t-1}{}_G - \alpha \nabla\mathcal{L}(\boldsymbol{w}^{t-1}{}_G) - \boldsymbol{w}^*\| + \alpha\|\nabla\mathcal{L}(\boldsymbol{w}^{t-1}{}_G) - \boldsymbol{g}_G^{t-1}\|. \quad (21)
\end{aligned}$$

Next, we expand the right-hand term of inequity (21) using Lemma 1–4. The detailed proofs are presented below.

Proof of Lemma 1: Repeatedly using the triangle inequality, we have:

$$\begin{aligned}
&\|\boldsymbol{g}_G^t - \nabla\mathcal{L}(\boldsymbol{w}^t{}_G)\| = \Big\| \sum_{i\in\mathbb{S}^t} \hat{q}_i^t \frac{\|\boldsymbol{g}_R^t\|}{\|\boldsymbol{g}_i^t\|} \boldsymbol{g}_i^t - \nabla\mathcal{L}(\boldsymbol{w}^t{}_G)\Big\| \\
&= \Big\| \sum_{i\in\mathbb{S}^t} \hat{q}_i^t \frac{\|\boldsymbol{g}_R^t\|}{\|\boldsymbol{g}_i^t\|} \boldsymbol{g}_i^t - \boldsymbol{g}_R^t + \boldsymbol{g}_R^t - \nabla\mathcal{L}(\boldsymbol{w}^t{}_G)\Big\| \\
&\le \Big\| \sum_{i\in\mathbb{S}^t} \hat{q}_i^t \frac{\|\boldsymbol{g}_R^t\|}{\|\boldsymbol{g}_i^t\|} \boldsymbol{g}_i^t - \boldsymbol{g}_R^t\Big\| + \|\boldsymbol{g}_R^t - \nabla\mathcal{L}(\boldsymbol{w}^t{}_G)\| \\
&\le \Big\| \sum_{i\in\mathbb{S}^t} \hat{q}_i^t \frac{\|\boldsymbol{g}_R^t\|}{\|\boldsymbol{g}_i^t\|} \boldsymbol{g}_i^t\Big\| + \|-\boldsymbol{g}_R^t\| + \|\boldsymbol{g}_R^t - \nabla\mathcal{L}(\boldsymbol{w}^t{}_G)\| \\
&\le \sum_{i\in\mathbb{S}^t} \hat{q}_i^t \Big\| \frac{\|\boldsymbol{g}_R^t\|}{\|\boldsymbol{g}_i^t\|} \boldsymbol{g}_i^t\Big\| + \|\boldsymbol{g}_R^t\| + \|\boldsymbol{g}_R^t - \nabla\mathcal{L}(\boldsymbol{w}^t{}_G)\| \\
&\overset{(22.1)}{=} \sum_{i\in\mathbb{S}^t} \hat{q}_i^t \|\boldsymbol{g}_R^t\| + \|\boldsymbol{g}_R^t\| + \|\boldsymbol{g}_R^t - \nabla\mathcal{L}(\boldsymbol{w}^t{}_G)\| \\
&= \Big(1 + \sum_{i\in\mathbb{S}^t} \hat{q}_i^t\Big) \|\boldsymbol{g}_R^t\| + \|\boldsymbol{g}_R^t - \nabla\mathcal{L}(\boldsymbol{w}^t{}_G)\| = \Big(1 + \\
&\sum_{i\in\mathbb{S}^t} \hat{q}_i^t\Big)\|\boldsymbol{g}_R^t - \nabla\mathcal{L}(\boldsymbol{w}^t{}_G) + \nabla\mathcal{L}(\boldsymbol{w}^t{}_G)\| + \|\boldsymbol{g}_R^t - \nabla\mathcal{L}(\boldsymbol{w}^t{}_G)\| \\
&\le \Big(2 + \sum_{i\in\mathbb{S}^t} \hat{q}_i^t\Big) \|\boldsymbol{g}_R^t - \nabla\mathcal{L}(\boldsymbol{w}^t{}_G)\| + \Big(1 + \sum_{i\in\mathbb{S}^t} \hat{q}_i^t\Big) \|\nabla\mathcal{L}(\boldsymbol{w}^t{}_G)\| \quad (22)
\end{aligned}$$

where inequality (22.1) is because $\frac{\boldsymbol{g}_i^t}{\|\boldsymbol{g}_i^t\|}$ is a unit vector. □

Proof of Lemma 2: The proof of Lemma 2 needs to rely on Assumption 2. And it has been proved in [26]. □

Proof of Lemma 3: First, we prove Eq. (18) of Lemma 3. Let $\boldsymbol{V} = \left\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{N_{\frac{1}{2}}}\right\}$ be an $\frac{1}{2}$-cover of the unit sphere $\boldsymbol{B}$. It is shown in Lemma 5.2, Lemma 5.3 [67], and Lemma 3.6 in [51] that $N_{\frac{1}{2}} \le d\log 6$, and

$$\begin{aligned}
&\Big\| \frac{1}{|D_R{}^t|} \sum_{X\in D_R{}^t} \nabla L(X, \boldsymbol{w}^*) - \nabla\mathcal{L}(\boldsymbol{w}^*)\Big\| \\
&\le 2\sup_{\boldsymbol{v}\in\boldsymbol{V}} \left\{ \Big\langle \frac{1}{|D_R{}^t|} \sum_{X\in D_R{}^t} \nabla L(X, \boldsymbol{w}^*) - \nabla\mathcal{L}(\boldsymbol{w}^*), \boldsymbol{v}\Big\rangle \right\}. \quad (23)
\end{aligned}$$

By the concentration inequalities for sub-exponential random variables introduced in the Lemma 3 [26], when Assumption 1 and $\Delta_1 \le \sigma_1^2/\gamma_1$ hold, we have:

$$\begin{aligned}
&\mathbb{P}\Big\{\Big\langle \frac{1}{|D_R{}^t|} \sum_{X\in D_R{}^t} \nabla L(X, \boldsymbol{w}^*) - \nabla\mathcal{L}(\boldsymbol{w}^*), \boldsymbol{v}\Big\rangle \\
&\ge (\Delta_1 + \mathcal{B}(\pi))\Big\} \le \exp(-|D_R{}^t|\Delta_1^2/(2\sigma_1^2)) \quad (24)
\end{aligned}$$

By taking the union bound over all elements in $\boldsymbol{V}$ and combining them with of Eq. (23) and Eq. (24), the following inequality is always true:

$$\begin{aligned}
&\mathbb{P}\left\{ \Big\| \frac{1}{|D_R{}^t|} \sum_{X\in D_R{}^t} \nabla L(X, \boldsymbol{w}^*) - \nabla\mathcal{L}(\boldsymbol{w}^*)\Big\| \ge 2\Delta_1 + 2\mathcal{B}(\pi)\right\} \\
&\le \exp(-|D_R{}^t|\Delta_1^2/(2\sigma_1^2)) = \exp(-\log(3/\delta)) = \frac{\delta}{3}. \quad (25)
\end{aligned}$$

Then, we prove Eq. (19) of Lemma 3 in a similar way.

$$\begin{aligned}
&\Big\| \frac{1}{|D_R{}^t|} \sum_{X\in D_R{}^t} \nabla h(X, \boldsymbol{w}^*) - \mathbb{E}_{X\sim\mathcal{X}}[h(X, \boldsymbol{w})]\Big\| \le 2\|\boldsymbol{w} - \\
&\boldsymbol{w}^*\| \sup_{\boldsymbol{v}\in\boldsymbol{V}} \left\{ \Big\langle \frac{1}{|D_R{}^t|} \sum_{X\in D_R{}^t} \nabla h(X, \boldsymbol{w}^*) - \mathbb{E}_{X\sim\mathcal{X}}[h(X, \boldsymbol{w})], \boldsymbol{v}\Big\rangle \right\} \quad (26)
\end{aligned}$$

and

$$\begin{aligned}
&\mathbb{P}\Big\{\Big\langle \frac{1}{|D_R{}^t|} \sum_{X\in D_R{}^t} \nabla h(X, \boldsymbol{w}^*) - \mathbb{E}_{X\sim\mathcal{X}}[h(X, \boldsymbol{w})], \boldsymbol{v}\Big\rangle \\
&\ge (\Delta_3 + \mathcal{B}(\pi))\Big\} \le \exp(-|D_R{}^t|\Delta_3^2/(2\sigma_2^2)). \quad (27)
\end{aligned}$$

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By taking the union bound over all elements in $\boldsymbol{V}$ and combining them with of Eq. (26) and Eq. (27), the following inequality is always true:

$$\mathbb{P}\Big\{\Big\|\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla h(X,\boldsymbol{w}^*)-\mathbb{E}_{X\sim\mathcal{X}}[h(X,\boldsymbol{w})]\Big\|\geq 2\|\boldsymbol{w}-\boldsymbol{w}^*\|(\Delta_3+\mathcal{B}(\pi))\Big\}\leq\exp(-|D_R{}^t|\Delta_3^2/(2\sigma_2^2))=\frac{\delta}{3}. \tag{28}$$

□

Proof of Lemma 4: The proof of Lemma 4 is mainly based on the ϵ-net argument [67] and theoretical results in the previous work [26], [51]. First, the joint training dataset of all clients in $\mathbb{S}_R^t$ is denoted as $D_R{}^t=\bigcup_{i\in\mathbb{S}_R^t}D_i$. We use the classical ε-net argument to find an intermediate value that connects $\boldsymbol{g}_R^t=\frac{1}{|D_R{}^t|}\sum_{X_i\in D_R{}^t}\nabla L(X_i,\boldsymbol{w})$ and $\nabla\mathcal{L}(\boldsymbol{w})$. Let $\tau=\frac{\gamma_2\sigma_1}{2\sigma_2^2}\sqrt{\frac{d}{|D_R{}^t|}}$, and $\mathscr{E}^*$ be an integer satisfying $\mathscr{E}^*=\lceil r\sqrt{d}/\tau\rceil$, where $\lceil\cdot\rceil$ is the upward rounding function. For integers $1\leq\mathscr{E}\leq\mathscr{E}^*$, we define a parameter set as $\Omega_{\mathscr{E}}=\{\boldsymbol{w}:\|\boldsymbol{w}-\boldsymbol{w}^*\|\leq\tau\mathscr{E}\}$. For a given integer $\mathscr{E}$, let $\boldsymbol{w}_1,\boldsymbol{w}_2,\ldots,\boldsymbol{w}_{N_{\varepsilon_{\mathscr{E}}}}$ be an $\varepsilon_{\mathscr{E}}$-cover of $\Omega_{\mathscr{E}}$, where $\varepsilon_{\mathscr{E}}=\frac{\sigma_2\tau\mathscr{E}}{\mathcal{K}_2}\sqrt{\frac{d}{|D_R{}^t|}}$ and $\mathcal{K}_2=\max\{\mathcal{K},\mathcal{K}_1\}$. By Lemma 5.2 in [67], $\log N_{\varepsilon_{\mathscr{E}}}\leq d\log\left(\frac{3\tau\mathscr{E}}{\varepsilon_{\mathscr{E}}}\right)$. Fix any $\boldsymbol{w}\in\Omega_{\mathscr{E}}$, there exists a $1\leq j_{\mathscr{E}}\leq N_{\varepsilon_{\mathscr{E}}}$ such that:

$$\|\boldsymbol{w}-\boldsymbol{w}_{j_{\mathscr{E}}}\|\leq\varepsilon_{\mathscr{E}}. \tag{29}$$

Then, by triangle's inequality, $\|\boldsymbol{g}_R^t-\nabla\mathcal{L}(\boldsymbol{w}^t{}_G)\|$ is divided into three parts,

$$\begin{aligned}\|\boldsymbol{g}_R^t-\nabla\mathcal{L}(\boldsymbol{w}^t{}_G)\|&=\Big\|\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla L(X,\boldsymbol{w})-\nabla\mathcal{L}(\boldsymbol{w})\Big\|\\&=\Big\|(\nabla\mathcal{L}(\boldsymbol{w}_{j_{\mathscr{E}}})-\nabla\mathcal{L}(\boldsymbol{w}))+\Big(\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla L(X,\boldsymbol{w})\\&\quad-\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla L(X,\boldsymbol{w}_{j_{\mathscr{E}}})\Big)+\Big(\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla L(X,\boldsymbol{w}_{j_{\mathscr{E}}})\\&\quad-\nabla\mathcal{L}(\boldsymbol{w}_{j_{\mathscr{E}}})\Big)\Big\|\leq\underbrace{\|\nabla\mathcal{L}(\boldsymbol{w})-\nabla\mathcal{L}(\boldsymbol{w}_{j_{\mathscr{E}}})\|}_{(30.A)}\\&\quad+\underbrace{\Big\|\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla L(X,\boldsymbol{w})-\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla L(X,\boldsymbol{w}_{j_{\mathscr{E}}})\Big\|}_{(30.B)}\\&\quad+\underbrace{\Big\|\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla L(X,\boldsymbol{w}_{j_{\mathscr{E}}})-\nabla\mathcal{L}(\boldsymbol{w}_{j_{\mathscr{E}}})\Big\|}_{(30.C)}\end{aligned} \tag{30}$$

In view of Assumption 2 and Eq.(29), (30.A) satisfies:

$$\|\nabla\mathcal{L}(\boldsymbol{w})-\nabla\mathcal{L}(\boldsymbol{w}_{j_{\mathscr{E}}})\|\leq\mathcal{K}\|\boldsymbol{w}-\boldsymbol{w}_{j_{\mathscr{E}}}\|\leq\mathcal{K}\varepsilon_{\mathscr{E}} \tag{31}$$

To analyze (30.B), we define an event E_1 as follows:

$$E_1=\left\{\sup_{\boldsymbol{w},\boldsymbol{w}'\in\Omega:\boldsymbol{w}\neq\boldsymbol{w}'}\frac{\|\nabla\hat{L}(D_R{}^t,\boldsymbol{w})-\nabla\hat{L}(D_R{}^t,\boldsymbol{w}')\|}{\|\boldsymbol{w}-\boldsymbol{w}'\|}\leq|D_R{}^t|\mathcal{K}_1\right\}, \tag{32}$$

where $\nabla\hat{L}(D_R{}^t,\boldsymbol{w})=\sum_{X_i\in D_R{}^t}\nabla L(X_i,\boldsymbol{w})$.

By Assumption 2, we have$\mathbb{P}\{E_1\}\geq1-\frac{\delta}{3}$. On E_1, we have:

$$\sup_{\boldsymbol{w}\in\Omega}\|\nabla\hat{L}(D_R{}^t,\boldsymbol{w})-\nabla\hat{L}(D_R{}^t,\boldsymbol{w}_{j_{\mathscr{E}}})\|\leq|D_R{}^t|\mathcal{K}_1\|\boldsymbol{w}-\boldsymbol{w}_{j_{\mathscr{E}}}\|\leq|D_R{}^t|\mathcal{K}_1\varepsilon_{\mathscr{E}}. \tag{33}$$

By triangle's inequality, (30.C) is scaled up as follows:

$$\begin{aligned}&\Big\|\frac{1}{|D_R{}^t|}\sum_{X_i\in D_R{}^t}\nabla L(X,\boldsymbol{w}_{j_{\mathscr{E}}})-\nabla\mathcal{L}(\boldsymbol{w}_{j_{\mathscr{E}}})\Big\|\\&=\Big\|\left(\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla L(X,\boldsymbol{w}^*)-\nabla\mathcal{L}(\boldsymbol{w}^*)\right)+\\&\left(\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}(\nabla L(X,\boldsymbol{w}_{j_{\mathscr{E}}})-\nabla L(X,\boldsymbol{w}^*))\right)+\\&\left(\nabla\mathcal{L}(\boldsymbol{w}^*)-\nabla\mathcal{L}(\boldsymbol{w}_{j_{\mathscr{E}}})\right)\Big\|\\&\overset{(34.1)}{\leq}\Big\|\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla L(X,\boldsymbol{w}^*)-\nabla\mathcal{L}(\boldsymbol{w}^*)\Big\|+\\&\Big\|\frac{1}{|D_R{}^t|}h(X,\boldsymbol{w}_{j_{\mathscr{E}}})-\mathbb{E}_{X\sim\mathcal{X}}[h(X,\boldsymbol{w}_{j_{\mathscr{E}}})]\Big\|\end{aligned} \tag{34}$$

where the term (34.1) is due to:

$$\begin{aligned}&\mathbb{E}_{X\sim\mathcal{X}}[h(X,\boldsymbol{w}_{j_{\mathscr{E}}})]=\mathbb{E}_{X\sim\mathcal{X}}[\nabla L(X,\boldsymbol{w}_{j_{\mathscr{E}}})-\nabla L(X,\boldsymbol{w}^*)]\\&=\mathbb{E}_{X\sim\mathcal{X}}[\nabla L(X,\boldsymbol{w}_{j_{\mathscr{E}}})]-\mathbb{E}_{X\sim\mathcal{X}}[\nabla L(X,\boldsymbol{w}^*)]=\\&\nabla\mathcal{L}(\boldsymbol{w}_{j_{\mathscr{E}}})-\nabla\mathcal{L}(\boldsymbol{w}^*).\end{aligned} \tag{35}$$

To analyze (30.C), we define events E_2 and E_3 as follows:

$$E_2=\left\{\Big\|\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla L(X,\boldsymbol{w}^*)-\nabla\mathcal{L}(\boldsymbol{w}^*)\Big\|\leq2\Delta_1+2\mathcal{B}(\pi)\right\} \tag{36}$$

$$E_3(\mathscr{E})=\Big\{\sup_{1\leq j\leq N_{\varepsilon_{\mathscr{E}}}}\Big\|\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}h(X,\boldsymbol{w}_j)-\mathbb{E}_{X\sim\mathcal{X}}[h(X,\boldsymbol{w}_j)]\Big\|\leq2(\Delta_2+\mathcal{B}(\pi))\tau\mathscr{E}\Big\}. \tag{37}$$

By Lemma 3 in this paper, Proposition 3.8 in [51], $\Delta_1\leq\sigma_1^2/\gamma_1$ and $\Delta_2\leq\sigma_2^2/\gamma_2$ (given in Lemma 4), we have $\mathbb{P}\{E_2\}\geq1-\frac{\delta}{3}$ and $\mathbb{P}\{E_3(\mathscr{E})\}\geq1-\frac{\delta}{3\mathscr{E}^*}$.

On event $E_1\cap E_2\cap E_3(\mathscr{E})$, we have:

$$\begin{aligned}&\sup_{\boldsymbol{w}\in\Omega_{\mathscr{E}}}\Big\|\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla L(X,\boldsymbol{w})-\nabla\mathcal{L}(\boldsymbol{w})\Big\|\leq\\&\underbrace{\mathcal{K}\varepsilon_{\mathscr{E}}}_{(38.A)}+\underbrace{\mathcal{K}_1\varepsilon_{\mathscr{E}}}_{(38.B)}+\underbrace{2\Delta_1+2\mathcal{B}(\pi)}_{(38.C1)}+\underbrace{2(\Delta_2+\mathcal{B}(\pi))\tau\mathscr{E}}_{(38.C2)}\\&\overset{(38.1)}{\leq}2\mathcal{K}_2\varepsilon_{\mathscr{E}}+2\Delta_1+2\mathcal{B}(\pi)+2\Delta_2\tau\mathscr{E}+2\mathcal{B}(\pi)\tau\mathscr{E}\overset{(38.2)}{\leq}\\&6\Delta_2\tau\mathscr{E}+2\Delta_1+2\Delta_2\end{aligned} \tag{38}$$

where the inequality can be split into (38.A), (38.B), (38.C1) and (38.C2) due to Eq. (31), (33), (25) and (28), respectively. And the inequality (38.1) holds because $(\mathcal{K}+\mathcal{K}_1)\leq2\mathcal{K}_2$. And the inequality (38.2) holds because $\Delta_2\geq\sigma_2\sqrt{\frac{d}{|D_R{}^t|}}$ and $\mathcal{B}(\pi)\leq\Delta_2$. According to the union bound, $\mathbb{P}\{E_1\cap E_2\cap(\cap_{\mathscr{E}=1}^{\mathscr{E}^*}E_3(\mathscr{E}))\}\geq$

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$1-\delta$ holds. Then, for any $\boldsymbol{w} \in \Omega_{\mathscr{E}^*}$, there exists an $1 \leq \mathscr{E} \leq \mathscr{E}^*$ such that $(\mathscr{E}-1)\tau < \|\boldsymbol{w}-\boldsymbol{w}^*\| \leq \mathscr{E}\tau$ holds. If $\mathscr{E}=1$, then

$$\begin{aligned}&\left\|\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla L(X,\boldsymbol{w})-\nabla\mathcal{L}(\boldsymbol{w})\right\|\\&\leq 6\Delta_2\tau+2\Delta_1+2\Delta_2\leq 5\Delta_1+2\Delta_2.\end{aligned}\tag{39}$$

where it holds due to $\Delta_2 \leq \frac{\sigma_2^2}{\gamma_2}$, $\Delta_1 \geq \sigma_1\sqrt{\frac{d}{|D_R{}^t|}}$.

$$\begin{aligned}6\Delta_2\tau &= 6\Delta_2\frac{\gamma_2\sigma_1}{2\sigma_2^2}\sqrt{\frac{d}{|D_R{}^t|}} = 3\Delta_2\frac{\gamma_2\sigma_1}{\sigma_2^2}\sqrt{\frac{d}{|D_R{}^t|}}\\&\leq 3\frac{\sigma_2^2}{\gamma_2}\frac{\gamma_2\sigma_1}{\sigma_2^2}\sqrt{\frac{d}{|D_R{}^t|}} = 3\sigma_1\sqrt{\frac{d}{|D_R{}^t|}}\leq 3\Delta_1.\end{aligned}\tag{40}$$

If $\mathscr{E} \geq 2$, then we have $2(\mathscr{E}-1) \geq \mathscr{E}$ and the following inequality holds:

$$\begin{aligned}&\left\|\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla L(X,\boldsymbol{w})-\nabla\mathcal{L}(\boldsymbol{w})\right\|\leq\\&6\Delta_2\tau(2(\mathscr{E}-1))+2\Delta_1+2\Delta_2 = 12\Delta_2\tau(\mathscr{E}-1)+\\&2\Delta_1+2\Delta_2\leq 12\Delta_2\|\boldsymbol{w}-\boldsymbol{w}^*\|+2\Delta_1+2\Delta_2.\end{aligned}\tag{41}$$

By combining Eq. (39) and Eq. (41), we have:

$$\begin{aligned}&\sup_{\boldsymbol{w}\in\Omega_{\mathscr{E}^*}}\left\|\frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla L(X,\boldsymbol{w})-\nabla\mathcal{L}(\boldsymbol{w})\right\|\leq\\&12\Delta_2\|\boldsymbol{w}-\boldsymbol{w}^*\|+5\Delta_1+2\Delta_2.\end{aligned}\tag{42}$$

Finally, since $\Omega \subset \Omega_{\mathscr{E}^*}$, we conclude the proof of Lemma 4, where $\boldsymbol{g}_R^t = \frac{1}{|D_R{}^t|}\sum_{X\in D_R{}^t}\nabla L(X,\boldsymbol{w}^t)$. □

Based on inequity (21) and Lemma 1 to Lemma 4, for any t^{th} global training iteration, we have:

$$\begin{aligned}&\|\boldsymbol{w}^t{}_G-\boldsymbol{w}^*\|\leq\|\boldsymbol{w}^{t-1}{}_G-\alpha\nabla\mathcal{L}(\boldsymbol{w}^{t-1}{}_G)-\boldsymbol{w}^*\|+\\&\alpha\|\nabla\mathcal{L}(\boldsymbol{w}^{t-1}{}_G)-\boldsymbol{g}_G^{t-1}\|\\&\overset{(43.1)}{\leq}\|\boldsymbol{w}^{t-1}{}_G-\alpha\nabla\mathcal{L}(\boldsymbol{w}^{t-1}{}_G)\\&\quad-\boldsymbol{w}^*\|+\alpha\left(1+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\right)\|\nabla\mathcal{L}(\boldsymbol{w}^{t-1}{}_G)\|\\&\quad+\alpha\left(2+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\right)\|\boldsymbol{g}_R^{t-1}-\nabla\mathcal{L}(\boldsymbol{w}^{t-1}{}_G)\|\\&\overset{(43.2)}{=}\underbrace{\|\boldsymbol{w}^{t-1}{}_G-\alpha\nabla\mathcal{L}(\boldsymbol{w}^{t-1}{}_G)-\boldsymbol{w}^*\|}_{(43.A)}+\\&\underbrace{\alpha\left(2+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\right)\|\boldsymbol{g}_R^{t-1}-\nabla\mathcal{L}(\boldsymbol{w}^{t-1}{}_G)\|}_{(43.B)}\\&\underbrace{+\alpha\left(1+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\right)\|\nabla\mathcal{L}(\boldsymbol{w}^{t-1}{}_G-\nabla\mathcal{L}(\boldsymbol{w}^*))\|}_{(43.C)}\\&\overset{(43.3)}{\leq}\underbrace{\sqrt{1-\frac{\mu^2}{4\mathcal{K}^2}}\|\boldsymbol{w}^{t-1}{}_G-\boldsymbol{w}^*\|}_{(43.D)}\\&\underbrace{+\alpha\left(2+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\right)(12\Delta_2\|\boldsymbol{w}^{t-1}{}_G-\boldsymbol{w}^*\|+5\Delta_1+2\Delta_2)}_{(43.E)}\\&\underbrace{+\alpha\left(1+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\right)\mathcal{K}\|\boldsymbol{w}^{t-1}{}_G-\boldsymbol{w}^*\|}_{(43.F)}\\&=\Bigg(\sqrt{1-\frac{\mu^2}{4\mathcal{K}^2}}+12\Delta_2\alpha\left(2+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\right)+\\&\alpha\left(1+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\right)\mathcal{K}\Bigg)\|\boldsymbol{w}^{t-1}{}_G-\boldsymbol{w}^*\|\\&+\alpha\left(2+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\right)(5\Delta_1+2\Delta_2)\end{aligned}\tag{43}$$

where the inequality (43.1) is obtained based on Lemma 1; we add $\nabla\mathcal{L}(\boldsymbol{w}^*)$ in (43.2) since $\nabla\mathcal{L}(\boldsymbol{w}^*)=0$; Eq.(43.A) scales to the term (43.D) via Eq.(17) in Lemma 2; The term (43.B) scales to the term (43.E) due to Lemma 4; The term (43.C) scales to the one (43.F) because of Assumption 2.

By recursively applying Eq.(43) for each global iteration, we have:

$$\begin{aligned}&\|\boldsymbol{w}^t{}_G-\boldsymbol{w}^*\|\leq(1-\rho)^t\|\boldsymbol{w}^0{}_G-\boldsymbol{w}^*\|\\&+\frac{\alpha\left(2+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\right)(5\Delta_1+2\Delta_2)}{\rho},\end{aligned}\tag{44}$$

where $\rho = 1-\left(\sqrt{1-\frac{\mu^2}{4\mathcal{K}^2}}+12\Delta_2\alpha\left(2+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\right)+\alpha\left(1+\sum_{i\in\mathbb{S}^t}\hat{q}_i^t\right)\mathcal{K}\right)$.

Finally, we conclude the proof of Theorem 1.

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